

Package ‘ICAMS’

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Type Package

Title In-Depth Characterization and Analysis of Mutational Signatures ('ICAMS')

Version 3.0.12

Description Analysis and visualization of experimentally elucidated mutational signatures -- the kind of analysis and visualization in Boot et al.,
``In-depth characterization of the cisplatin mutational signature in human cell lines and in esophageal and liver tumors", Genome Research 2018, <doi:10.1101/gr.230219.117> and
``Characterization of colibactin-associated mutational signature in an Asian oral squamous cell carcinoma and in other mucosal tumor types", Genome Research 2020 <doi:10.1101/gr.255620.119>.
'ICAMS' stands for In-depth Characterization and Analysis of Mutational Signatures. 'ICAMS' has functions to read in variant call files (VCFs) and to collate the corresponding catalogs of mutational spectra and to analyze and plot catalogs of mutational spectra and signatures. Handles both ``counts-based" and ``density-based" (i.e. representation as mutations per megabase) mutational spectra or signatures.

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URL <https://github.com/steverozen/ICAMS>

BugReports <https://github.com/steverozen/ICAMS/issues>

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ICAMS-package

ICAMS: In-depth Characterization and Analysis of Mutational Signatures

Description

Analysis and visualization of experimentally elucidated mutational signatures – the kind of analysis and visualization in Boot et al., "In-depth characterization of the cisplatin mutational signature in human cell lines and in esophageal and liver tumors",

Genome Research 2018 <https://doi.org/10.1101/gr.230219.117> and "Characterization of colibactin-associated mutational signature in an Asian oral squamous cell carcinoma and in other mucosal tumor types", *Genome Research* 2020, <https://doi.org/10.1101/gr.255620.119>. "ICAMS" stands for In-depth Characterization and Analysis of Mutational Signatures. "ICAMS" has functions to read in variant call files (VCFs) and to collate the corresponding catalogs of mutational spectra and to analyze and plot catalogs of mutational spectra and signatures.

Details

"ICAMS" can read in VCFs generated by Strelka, Mutect or other variant callers, and collate the mutations into "catalogs" of mutational spectra. "ICAMS" can create and plot catalogs of mutational spectra or signatures for single base substitutions (SBS), doublet base substitutions (DBS), and small insertions and deletions (ID). It can also read and write these catalogs.

Catalogs

A key data type in "ICAMS" is a "catalog" of mutation counts, of mutation densities (see below), or of mutational signatures.

Catalogs are S3 objects of class `matrix` and one of several additional classes that specify the types of the mutations represented in the catalog. The additional class is one of

- `SBS96Catalog` (strand-agnostic single base substitutions in trinucleotide context)
- `SBS192Catalog` (transcription-stranded single-base substitutions in trinucleotide context)
- `SBS1536Catalog`
- `DBS78Catalog`
- `DBS144Catalog`
- `DBS136Catalog`
- `IndelCatalog`
- `ID166Catalog` (genic-intergenic indel catalog)

[as.catalog](#) is the main constructor.

Conceptually, a catalog also has one of the following types, indicated by the attribute `catalog.type`:

1. Matrix of mutation counts (one column per sample), representing (counts-based) mutational spectra (`catalog.type = "counts"`).
2. Matrix of mutation **densities**, i.e. mutations per occurrences of source sequences (one column per sample), representing (density-based) mutational spectra (`catalog.type = "density"`).
3. Matrix of mutational signatures, which are similar to spectra. However where spectra consist of counts or densities of mutations in each mutation class (e.g. `ACA > AAA`, `ACA > AGA`, `ACA > ATA`, `ACC > AAC`, ...), signatures consist of the proportions of mutations in each class (with all the proportions summing to 1). A mutational signature can be based on either:
 - mutation counts (a "counts-based mutational signature", `catalog.type = "counts.signature"`),
 - or
 - mutation densities (a "density-based mutational signature", `catalog.type = "density.signature"`).

Catalogs also have the attribute `abundance`, which contains the counts of different source sequences for mutations. For example, for SBSs in trinucleotide context, the abundances would be the counts of each trinucleotide in the human genome, exome, or in the transcribed region of the genome. See [TransformCatalog](#) for more information. Abundances logically depend on the species in question and on the part of the genome being analyzed.

In "ICAMS" abundances can sometimes be inferred from the catalog class attribute and the function arguments `region`, `ref.genome`, and `catalog.type`. Otherwise abundances can be provided as an abundance argument. See [all.abundance](#) for examples.

Possible values for `region` are the strings `genome`, `transcript`, `exome`, and `unknown`; `transcript` includes entire transcribed regions, i.e. the introns as well as the exons.

If you need to create a catalog from a source other than this package (i.e. other than with [ReadCatalog](#) or [VCFsToCatalogs](#), [VCFsToZipFile](#), etc.), then use [as.catalog](#).

Subscripting catalogs

If user wants to subscript specific columns from a catalog, it is needed to call `library(ICAMS)` beforehand to preserve the ICAMS catalog attribute. Otherwise writing or plotting catalog function in ICAMS may not work properly.

Creating catalogs from variant call files (VCF files)

- [VCFsToCatalogs](#) creates 3 SBS catalogs (96, 192, 1536), 3 DBS catalogs (78, 136, 144) and ID (small insertions and deletions) catalog from the VCFs.

Plotting catalogs

- [PlotCatalog](#) function plots mutational spectra for **one** sample or plot **one** mutational signature.
- [PlotCatalogToPdf](#) function plots catalogs of mutational spectra or of mutational signatures to a PDF file.

Wrapper function to create catalogs from VCFs and plot the catalogs to PDF files

- [VCFsToCatalogsAndPlotToPdf](#) creates all types of SBS, DBS and ID catalogs from VCFs and plots the catalogs.

Wrapper function to create a zip file which contains catalogs and plot PDFs from VCF files

- [VCFsToZipFile](#) creates a zip file which contains SBS, DBS and ID catalogs and plot PDFs from VCF files.

The `ref.genome` (reference genome) argument

Many functions take the argument `ref.genome`.

To create a mutational spectrum catalog from a VCF file, "ICAMS" needs the reference genome sequence that matches the VCF file. The `ref.genome` argument provides this.

`ref.genome` must be one of

1. A variable from the Bioconductor [BSgenome](#) package that contains a particular reference genome, for example `BSgenome.Hsapiens.1000genomes.hs37d5`.
2. The strings "hg38" or "GRCh38", which specify `BSgenome.Hsapiens.UCSC.hg38`.
3. The strings "hg19" or "GRCh37", which specify `BSgenome.Hsapiens.1000genomes.hs37d5`.
4. The strings "mm10" or "GRCm38", which specify `BSgenome.Mmusculus.UCSC.mm10`.

All needed reference genomes must be installed separately by the user. Further instructions are at <https://bioconductor.org/packages/release/bioc/html/BSgenome.html>.

Use of "ICAMS" with reference genomes other than the 2 human genomes and 1 mouse genome specified above is restricted to `catalog.type` of counts or counts.signature unless the user also creates the necessary abundance vectors. See [all.abundance](#).

Use [available.genomes\(\)](#) to get the list of available genomes.

Writing catalogs to files

- [WriteCatalog](#) function writes a catalog to a file.

Reading catalogs

- [ReadCatalog](#) function reads a file that contains a catalog in standardized format.

Transforming catalogs

[TransformCatalog](#) function transforms catalogs of mutational spectra or signatures to account for differing abundances of the source sequence of the mutations in the genome.

For example, mutations from ACG are much rarer in the human genome than mutations from ACC simply because CG dinucleotides are rare in the genome. Consequently, there are two possible representations of mutational spectra or signatures. One representation is based on mutation counts as observed in a given genome or exome, and this approach is widely used, as, for example, at <https://cancer.sanger.ac.uk/signatures/>, which presents signatures based on observed mutation counts in the human genome. We call these "counts-based spectra" or "counts-based signatures".

Alternatively, mutational spectra or signatures can be represented as mutations per source sequence, for example the number of ACT > AGT mutations occurring at all ACT 3-mers in a genome. We call these "density-based spectra" or "density-based signatures".

This function can also transform spectra based on observed genome-wide counts to "density"-based catalogs. In density-based catalogs mutations are expressed as mutations per source sequences. For example, a density-based catalog represents the proportion of ACCs mutated to ATCs, the proportion of ACGs mutated to ATGs, etc. This is different from counts-based mutational spectra catalogs, which contain the number of ACC > ATC mutations, the number of ACG > ATG mutations, etc.

This function can also transform observed-count based spectra or signatures from genome to exome based counts, or between different species (since the abundances of source sequences vary between genome and exome and between species).

Collapsing catalogs

[CollapseCatalog](#) function

1. Takes a mutational spectrum or signature catalog that is based on a fined-grained set of features (for example, single-nucleotide substitutions in the context of the preceding and following 2 bases).
2. Collapses it to a catalog based on a coarser-grained set of features (for example, single-nucleotide substitutions in the context of the immediately preceding and following bases).

Data

1. [CatalogRowOrder](#) Standard order of rownames in a catalog. The rownames encode the type of each mutation. For example, for SBS96 catalogs, the rowname AGAT represents a mutation from AGA > ATA.
2. [TranscriptRanges](#) Transcript ranges and strand information for a particular reference genome.
3. [all.abundance](#) The counts of different source sequences for mutations.
4. [GeneExpressionData](#) Example gene expression data from two cell lines.

#'

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See Also

Useful links:

- <https://github.com/steverozen/ICAMS>
- Report bugs at <https://github.com/steverozen/ICAMS/issues>

all.abundance	<i>K-mer abundances</i>
---------------	-------------------------

Description

An R list with one element each for BSgenome.Hsapiens.1000genomes.hs37d5, BSgenome.Hsapiens.UCSC.hg38 and BSgenome.Mmusculus.UCSC.mm10. Each element is in turn a sub-list keyed by exome, transcript, and genome. Each element of the sub list is keyed by the number of rows in the catalog class (as a string, e.g. "78", not 78). The keys are: 78 (DBS78Catalog), 96 (SBS96Catalog), 136 (DBS136Catalog), 144 (DBS144Catalog), 192 (SBS192Catalog), and 1536 (SBS1536Catalog). So, for example to get the exome abundances for SBS96 catalogs for BSgenome.Hsapiens.UCSC.hg38 exomes one would reference
`all.abundance[["BSgenome.Hsapiens.UCSC.hg38"]][["exome"]][["96"]]`
or `all.abundance$BSgenome.Hsapiens.UCSC.hg38$exome$"96"`. The value of the abundance is an integer vector with the K-mers as names and each value being the count of that K-mer.

Usage

```
all.abundance
```

Format

See Description.

Examples

```
all.abundance$BSgenome.Hsapiens.UCSC.hg38$transcript$`144`
#      AA      AC      AG      AT      CA      CC ...
# 90769160 57156295 85738416 87552737 83479655 63267896 ...
# There are 90769160 AAs on the sense strands of transcripts in
# this genome.
```

AnnotateIDVCF	<i>Add sequence context and transcript information to an in-memory ID (insertion/deletion) VCF, and confirm that they match the given reference genome</i>
---------------	--

Description

Add sequence context and transcript information to an in-memory ID (insertion/deletion) VCF, and confirm that they match the given reference genome

Usage

```
AnnotateIDVCF(
  ID.vcf,
  ref.genome,
  flag.mismatches = 0,
  name.of.VCF = NULL,
  suppress.discarded.variants.warnings = TRUE,
  explain_indels = 1,
  context_width_multiplier = 20L
)
```

Arguments

ID.vcf	An in-memory ID (insertion/deletion) VCF as a data.frame. This function expects that there is a "context base" to the left, for example REF = ACG, ALT = A (deletion of CG) or REF = A, ALT = ACC (insertion of CC).
ref.genome	Can be a string or a BSgenome. If a string, it should be a well-known name for reference genome in BSgenome
flag.mismatches	Deprecated. If there are ID variants whose REF do not match the extracted sequence from ref.genome, the function will automatically discard these variants. See element discarded.variants in the return value for more details.
name.of.VCF	Name of the VCF file.
suppress.discarded.variants.warnings	If TRUE, do warn when variants that cannot be processed are discarded.
explain_indels	If TRUE generate message on stdout showing how the indel was categorized.
context_width_multiplier	Used to guess how much sequence on each side of an indel is needed to categorize it.

Value

A list of elements:

- annotated.vcf: The original VCF data frame with two new columns added to the input data frame:
 - seq.context: The sequence embedding the variant.
 - seq.context.width: The width of seq.context to the left.
- discarded.variants: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column discarded.reason for more details.

Examples

```
file <- c(system.file("extdata/Strelka-ID-vcf/",
                     "Strelka.ID.GRCh37.s1.vcf",
                     package = "ICAMS"))
ID.vcf <- ReadAndSplitVCFs(file, variant.caller = "strelka")$ID[[1]]
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  list <- AnnotateIDVCF(ID.vcf, ref.genome = "hg19")
  annotated.ID.vcf <- list$annotated.vcf}
```

AnnotateSBSVCF	<i>Add sequence context and transcript information to an in-memory SBS VCF</i>
----------------	--

Description

Add sequence context and transcript information to an in-memory SBS VCF

Usage

```
AnnotateSBSVCF(SBS.vcf, ref.genome, trans.ranges = NULL, name.of.VCF = NULL)
```

Arguments

SBS.vcf	An in-memory SBS VCF as a data.frame.
ref.genome	A ref.genome argument as described in ICAMS .
trans.ranges	Optional. If ref.genome specifies one of the BSgenome object 1. BSgenome.Hsapiens.1000genomes.hs37d5 2. BSgenome.Hsapiens.UCSC.hg38 3. BSgenome.Mmusculus.UCSC.mm10 then the function will infer trans.ranges automatically. Otherwise, user will need to provide the necessary trans.ranges. Please refer to TranscriptRanges for more details. If is.null(trans.ranges) do not add transcript range information.
name.of.VCF	Name of the VCF file.

Value

An in-memory SBS VCF as a data.table. This has been annotated with the sequence context (column name seq.21bases) and with transcript information in the form of a gene symbol (e.g. "TP53") and transcript strand. This information is in the columns trans.start.pos, trans.end.pos, trans.strand, trans.Ensembl.gene.ID and trans.gene.symbol in the output. These columns are not added if is.null(trans.ranges).

Examples

```
file <- c(system.file("extdata/Strelka-SBS-vcf",
                     "Strelka.SBS.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadAndSplitVCFs(file, variant.caller = "strelka")
SBS.vcf <- list.of.vcfs$SBS[[1]]
```

```
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  annotated.SBS.vcf <- AnnotateSBSVCF(SBS.vcf, ref.genome = "hg19",
                                     trans.ranges = trans.ranges.GRCh37)}
}
```

as.catalog	Create a catalog from a matrix, data.frame, or vector
------------	---

Description

Create a catalog from a matrix, data.frame, or vector

Usage

```
as.catalog(
  object,
  ref.genome = NULL,
  region = "unknown",
  catalog.type = "counts",
  abundance = NULL,
  infer.rownames = FALSE
)
```

Arguments

object	A numeric matrix, numeric data.frame, or vector. If a vector, converted to a 1-column matrix with rownames taken from the element names of the vector and with column name "Unknown". If argument infer.rownames is FALSE then this argument must have rownames to denote the mutation types. See CatalogRowOrder for more details.
ref.genome	A ref.genome argument as described in ICAMS .
region	A character string designating a region, one of genome, transcript, exome, unknown; see ICAMS . If the catalog type is a stranded catalog type (SBS192 or DBS144), region = "genome" will be silently converted to "transcript".
catalog.type	One of "counts", "density", "counts.signature", "density.signature".
abundance	If NULL, then inferred if ref.genome is one of the reference genomes known to ICAMS and region is not unknown. See ICAMS . The argument abundance should contain the counts of different source sequences for mutations in the same format as the numeric vectors in all.abundance .
infer.rownames	If TRUE, and object has no rownames, then assume the rows of object are in the correct order and add the rownames implied by the number of rows in object (e.g. rownames for SBS 192 if there are 192 rows). If TRUE, be sure the order of rows is correct .

Value

A catalog as described in [ICAMS](#).

Examples

```
# Create an SBS96 catalog with all mutation counts equal to 1.
object <- matrix(1, nrow = 96, ncol = 1,
                 dimnames = list(catalog.row.order$SBS96))
catSBS96 <- as.catalog(object)
```

Canonicalize1Del

*Given a deletion and its sequence context, categorize it***Description**

This function has been superseded by `categorize_1_justified_indel`, but is temporarily maintained for regression testing.

Usage

```
Canonicalize1Del(context, del.seq, pos, trace = 0)
```

Arguments

<code>context</code>	The deleted sequence plus ample surrounding sequence on each side (at least as long as <code>del.seq</code>).
<code>del.seq</code>	The deleted sequence in context.
<code>pos</code>	The position of <code>del.seq</code> in context.
<code>trace</code>	If > 0 , then generate messages tracing how the computation is carried out.

Details

This function is primarily for internal use, but we export it to document the underlying logic.

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on deletion mutation classification.

This function first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Value

A string that is the canonical representation of the given deletion type. Return NA and raise a warning if there is an un-normalized representation of the deletion of a repeat unit. See [FindDelMH](#) for details. (This seems to be very rare.)

Examples

```
Canonicalize1Del("xyAAAqr", del.seq = "A", pos = 3) # "DEL:T:1:2"
Canonicalize1Del("xyAAAqr", del.seq = "A", pos = 4) # "DEL:T:1:2"
Canonicalize1Del("xyAqr", del.seq = "A", pos = 3)   # "DEL:T:1:0"
```

CatalogRowOrder	<i>Standard order of row names in a catalog</i>
-----------------	---

Description

This data is designed for those who need to create their own catalogs from formats not supported by this package. The rownames denote the mutation types. For example, for SBS96 catalogs, the rowname AGAT represents a mutation from AGA > ATA.

Usage

```
catalog.row.order
```

Format

A list of character vectors indicating the standard orders of row names in catalogs.

An object of class `list` of length 9.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-row/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+. In ID83 catalogs, deletion repeat sizes range from 0 to 5.

Examples

```
catalog.row.order$SBS96
# "ACAA" "ACCA" "ACGA" "ACTA" "CCAA" "CCCA" "CCGA" "CCTA" ...
# There are altogether 96 row names to denote the mutation types
# in SBS96 catalog.
```

categorize_1_justified_indel	<i>Given a indel and its sequence context, categorize it</i>
------------------------------	--

Description

This function is primarily for internal use, but we export it to document the underlying logic.

Usage

```
categorize_1_justified_indel(context, ins_or_del, ins_or_del_seq, pos)
```

Arguments

context	The sequence surrounding the indel PRIOR to the insertion or deletion.
ins_or_del	A single character, with "i" denoting an insertion and "d" denotene an deletion.
ins_or_del_seq	The the sequence that was inserted or deleted.
pos	For deletions, the 1-based position of the start of the deleted sequence; for insertions, the position immediately to the right of where the insertion occurs.

Details

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on deletion mutation classification.

This function first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDeIMH](#)).

Value

A string that is the canonical representation of the given deletion type. Return NA and raise a warning if there is an un-normalized representation of the deletion of a repeat unit. See [FindDeIMH](#) for details. (This seems to be very rare.)

Examples

```
categorize_1_justified_indel("GGAAAGG", "d", ins_or_del_seq = "A", pos = 3) # "DEL:T:1:2"
categorize_1_justified_indel("GGAAAGG", "d", ins_or_del_seq = "A", pos = 4) # "DEL:T:1:2"
categorize_1_justified_indel("TTATT", "d", ins_or_del_seq = "A", pos = 3) # "DEL:T:1:0"
```

CollapseCatalog

"Collapse" a catalog

Description

1. Take a mutational spectrum or signature catalog that is based on a fined-grained set of features (for example, single-nucleotide substitutions in the context of the preceding and following 2 bases).
2. Collapse it to a catalog based on a coarser-grained set of features (for example, single-nucleotide substitutions in the context of the immediately preceding and following bases).

Collapse192CatalogTo96 Collapse an SBS 192 catalog to an SBS 96 catalog.

Collapse1536CatalogTo96 Collapse an SBS 1536 catalog to an SBS 96 catalog.

Collapse144CatalogTo78 Collapse a DBS 144 catalog to a DBS 78 catalog.

Usage

```
Collapse192CatalogTo96(catalog)

Collapse1536CatalogTo96(catalog)

Collapse144CatalogTo78(catalog)
```

Arguments

catalog A catalog as defined in [ICAMS](#).

Value

A catalog as defined in [ICAMS](#).

Examples

```
# Create an SBS192 catalog and collapse it to an SBS96 catalog
object <- matrix(1, nrow = 192, ncol = 1,
                 dimnames = list(catalog.row.order$SBS192))
catSBS192 <- as.catalog(object, region = "transcript")
catSBS96 <- Collapse192CatalogTo96(catSBS192)
```

ConvertCatalogToSigProfilerFormat

Covert an ICAMS Catalog to SigProfiler format

Description

Specially, the row orders in ICAMS internal format (see `ICAMS::catalog.row.order`) are converted to headers in SigProfiler format.

Usage

```
ConvertCatalogToSigProfilerFormat(input.catalog, file, sep = "\t")
```

Arguments

input.catalog Either a character string, in which case this is the path to a file containing a catalog in [ICAMS](#) format, or an in-memory [ICAMS](#) catalog.

file The path of the file to be written.

sep Separator to use in the output file.

Details

For SigProfiler formats, please see the links below for:

- SBS: <https://osf.io/s93d5/wiki/5.%20Output%20-%20SBS/>
- DBS: <https://osf.io/s93d5/wiki/5.%20Output%20-%20DBS/>
- ID: <https://osf.io/s93d5/wiki/5.%20Output%20-%20ID/>

Note

This function can only transform SBS96, SBS192, SBS1536, DBS78 and ID ICAMS catalog to SigProfiler format.

Examples

```
path <- system.file("extdata",
                    "strelka.regress.cat.sbs.96.csv",
                    package = "ICAMS")
catSBS96 <- ReadCatalog(path)
ConvertCatalogToSigProfilerFormat(input.catalog = catSBS96,
                                  file = file.path(tempdir(), "sigproCat.txt"))
```

FindDelMH

*Return the length of microhomology at a deletion***Description**

Return the length of microhomology at a deletion

Usage

```
FindDelMH(context, deleted.seq, pos, trace = 0, warn.cryptic = TRUE)
```

Arguments

context	The deleted sequence plus ample surrounding sequence on each side (at least as long as del . sequence).
deleted.seq	The deleted sequence in context.
pos	The position of del . sequence in context.
trace	If > 0, then generate various messages showing how the computation is carried out.
warn.cryptic	if TRUE generating a warning if there is a cryptic repeat (see the example).

Details

This function is primarily for internal use, but we export it to document the underlying logic.

Example:

GGCTAGTT aligned to GGCTAGAACTAGTT with a deletion represented as:

```
GGCTAGAACTAGTT
GG-----CTAGTT  GGCTAGTT  GG[CTAGAA]CTAGTT
                   ----    ----
```

Presumed repair mechanism leading to this:


```

    . . . .
GGCTAGAACTAGTT
CCGATCTTGATCAA

```

=>

```

    . . . .
GGCTAG      TT
CC      GATCAA
    . . . .

```

=>

```

GGCTAGTT
CCGATCAA

```

Variant-caller software can represent the same deletion in several different, but completely equivalent, ways.

```

GGC-----TAGTT  GGCTAGTT  GGC[TAGAAC]TAGTT
                        *  ---  *  ---

```

```

GGCT-----AGTT  GGCTAGTT  GGCT[AGAACT]AGTT
                        **  --  **  --

```

```

GGCTA-----GTT  GGCTAGTT  GGCTA[GAACTA]GTT
                        ***  -  ***  -

```

```

GGCTAG-----TT  GGCTAGTT  GGCTAG[AACTAG]TT
                        *****  *****

```

This function finds:

1. The maximum match of undeleted sequence to the left of the deletion that is identical to the right end of the deleted sequence, and
2. The maximum match of undeleted sequence to the right of the deletion that is identical to the left end of the deleted sequence.

The microhomology sequence is the concatenation of items (1) and (2).

Warning

A deletion in a *repeat* can also be represented in several different ways. A deletion in a repeat is abstractly equivalent to a deletion with microhomology that spans the entire deleted sequence. For example;

```

GACTAGCTAGTT
GACTA----GTT  GACTAGTT  GACTA[GCTA]GTT
                        ***  -***  -

```

is really a repeat

```
GACTAG----TT GACTAGTT GACTAG[CTAG]TT
          ****  ----

GACT----AGTT GACTAGTT GACT[AGCT]AGTT
          **  ----*  --
```

This function only flags these "cryptic repeats" with a -1 return; it does not figure out the repeat extent.

Value

The length of the maximum microhomology of `del` sequence in context.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Examples

```
# GAGAGG[CTAGAA]CTAGTT
#      ----  ----
FindDelMH("GGAGAGGCTAGAACTAGTTAAAA", "CTAGAA", 8, trace = 0) # 4

# A cryptic repeat
#
# TAAATTATTTATTAATTTATTG
# TAAATTA----TTAATTTATTG = TAAATTATTAATTTATTG
#
# equivalent to
#
# TAAATTATTTATTAATTTATTG
# TAAAT----TATTAATTTATTG = TAAATTATTAATTTATTG
#
# and
#
# TAAATTATTTATTAATTTATTG
# TAAA----TTATTAATTTATTG = TAAATTATTAATTTATTG

FindDelMH("TAAATTATTTATTAATTTATTG", "TTTA", 8, warn.cryptic = FALSE) # -1
```

FindMaxRepeatDel

Return the number of repeat units in which a deletion is embedded

Description

Return the number of repeat units in which a deletion is embedded

Usage

```
FindMaxRepeatDel(context, rep.unit.seq, pos)
```

Arguments

context	A string that embeds rep.unit.seq at position pos
rep.unit.seq	A substring of context at pos to pos + nchar(rep.unit.seq) - 1, which is the repeat unit sequence.
pos	The position of rep.unit.seq in context.

Details

This function is primarily for internal use, but we export it to document the underlying logic.

For example `FindMaxRepeatDel("xyaczt", "ac", 3)` returns 0.

If `substr(context, pos, pos + nchar(rep.unit.seq) - 1) != rep.unit.seq` then stop.

If this functions returns 0, then it is necessary to look for microhomology using the function [FindDelMH](#).

Warning

This function depends on the variant caller having "aligned" the deletion within the context of the repeat.

For example, a deletion of CAG in the repeat

```
GTCAGCAGCATGT
```

can have 3 "aligned" representations as follows:

```
CT---CAGCAGGT
CTCAG---CAGGT
CTCAGCAG---GT
```

In these cases this function will return 2. (Please not that the return value does not include the rep.uni.seq in the count.)

However, the same deletion can also have an "unaligned" representation, such as

```
CTCAGC---AGGT
```

(a deletion of AGC).

In this case this function will return 1 (a deletion of AGC in a 2-element repeat of AGC).

Value

The number of repeat units in which rep.unit.seq is embedded, not including the input rep.unit.seq in the count.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Examples

```
FindMaxRepeatDel("xyACACzt", "AC", 3) # 1
FindMaxRepeatDel("xyACACzt", "CA", 4) # 0
```

GeneExpressionData	<i>Example gene expression data from two cell lines</i>
--------------------	---

Description

This data is designed to be used as an example in function [PlotTransBiasGeneExp](#) and [PlotTransBiasGeneExpToPdf](#).

Usage

```
gene.expression.data.HepG2

gene.expression.data.MCF10A
```

Format

A [data.table](#) which contains the expression values of genes.

An object of class `data.table` (inherits from `data.frame`) with 57736 rows and 4 columns.

An object of class `data.table` (inherits from `data.frame`) with 57736 rows and 4 columns.

Examples

```
gene.expression.data.HepG2
# Ensembl.gene.ID gene.symbol counts      TPM
# ENSG000000000003      TSPAN6    6007 33.922648455
# ENSG000000000005      TNMD       0  0.000000000
# ENSG000000000419      DPM1    4441 61.669371091
# ENSG000000000457      SCYL3    1368  3.334619195
# ENSG000000000460  C1orf112     916  2.416263423
#           ...           ...     ...           ...
```

GetVAF	<i>Extract the VAFs (variant allele frequencies) and read depth information from a VCF file</i>
--------	---

Description

Extract the VAFs (variant allele frequencies) and read depth information from a VCF file

Usage

```
GetStrelkaVAF(vcf, name.of.VCF = NULL)

GetMutectVAF(vcf, name.of.VCF = NULL, tumor.col.name = NA)

GetFreebayesVAF(vcf, name.of.VCF = NULL)

GetPCAWGConsensusVAF(vcf, mc.cores = 1)
```

Arguments

<code>vcf</code>	An in-memory VCF data frame.
<code>name.of.VCF</code>	Name of the VCF file.
<code>tumor.col.name</code>	Optional. Only applicable to Mutect VCF. Name or index of the column in Mutect VCF which contains the tumor sample information. It must have quotation marks if specifying the column name. If <code>tumor.col.name</code> is equal to NA(default), this function will use the 10th column to calculate VAFs.
<code>mc.cores</code>	The number of cores to use. Not available on Windows unless <code>mc.cores = 1</code> .

Value

The original `vcf` with two additional columns added which contain the VAF(variant allele frequency) and read depth information.

Note

[GetPCAWGConsensusVAF](#) is analogous to [GetMutectVAF](#), calculating VAF and read depth from PCAWG7 consensus vcfs

Examples

```
file <- c(system.file("extdata/Strelka-SBS-vcf",
                     "Strelka.SBS.GRCh37.s1.vcf",
                     package = "ICAMS"))
MakeDataFrameFromVCF <- getFromNamespace("MakeDataFrameFromVCF", "ICAMS")
df <- MakeDataFrameFromVCF(file)
df1 <- GetStrelkaVAF(df)
```

IsICAMSCatalog

Check whether an R object contains one of the ICAMS catalog classes

Description

Check whether an R object contains one of the ICAMS catalog classes

Check whether an R object contains one of the ICAMS catalog classes

Usage

```
IsICAMSCatalog(object)

IsICAMSCatalog(object)
```

Arguments

object An R object.

Value

A logical value.

A logical value.

Examples

```
# Create a matrix with all values being 1
object <- matrix(1, nrow = 96, ncol = 1,
                 dimnames = list(catalog.row.order$SBS96))
IsICAMSCatalog(object) # FALSE

# Use as.catalog to add class attribute to object
catalog <- as.catalog(object)
IsICAMSCatalog(catalog) # TRUE

# Create a matrix with all values being 1
object <- matrix(1, nrow = 96, ncol = 1,
                 dimnames = list(catalog.row.order$SBS96))
IsICAMSCatalog(object) # FALSE

# Use as.catalog to add class attribute to object
catalog <- as.catalog(object)
IsICAMSCatalog(catalog) # TRUE
```

justify_id_vcf	<i>Add sequence context and transcript information to an in-memory ID (insertion/deletion) VCF, and confirm that they match the given reference genome</i>
----------------	--

Description

Add sequence context and transcript information to an in-memory ID (insertion/deletion) VCF, and confirm that they match the given reference genome

Usage

```
justify_id_vcf(
  ID.vcf,
  ref.genome,
  name.of.VCF = NULL,
  suppress.discarded.variants.warnings = TRUE,
  explain_indels = 1,
  context_width_multiplier = 20L
)
```

Arguments

<code>ID.vcf</code>	An in-memory ID (insertion/deletion) VCF as a <code>data.frame</code> . This function expects that there is a "context base" to the left, for example <code>REF = ACG</code> , <code>ALT = A</code> (deletion of CG) or <code>REF = A</code> , <code>ALT = ACC</code> (insertion of CC).
<code>ref.genome</code>	Can be a string or a <code>BSgenome</code> . If a string, it should be a well-known name for reference genome in <code>BSgenome</code>
<code>name.of.VCF</code>	Name of the VCF file.
<code>suppress.discarded.variants.warnings</code>	If <code>TRUE</code> , do warn when variants that cannot be processed are discarded.
<code>explain_indels</code>	If 0, do not explain, if 1, explain indels that were justified (via messages), if 2, generate messages for all indels.
<code>context_width_multiplier</code>	Used to guess how much sequence on each side of an indel is needed to categorize it.

Value

A list of elements:

- `annotated.vcf`: The original VCF data frame in which `POS`, `REF` and `ALT` with new columns added to the input data frame:
 - `seq.context`: The sequence embedding the variant.
 - `seq.context.width`: The width of `seq.context` to the left of the indel
 - `pos_shift`
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.

Examples

```
file <- c(system.file("tests/testthat/testdata/",
                     "vcf_to_test_justify_indels_in_id_vcf_with_contexts.csv",
                     package = "ICAMS"))

stopifnot(file.exists(file))

read.csv(file) |> dplyr::select(-seq.context.width, -seq.context) -> vcf_no_contexts

if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  justify_id_vcf(vcf_no_contexts, ref.genome = "hg19", explain_indels = 2)}

```

<code>justify_indel</code>	<i>Move the notional position of a deletion or insertion as far left as possible.</i>
----------------------------	---

Description

short_string should be generated by a deletion in long_str at (1-based) position pos. If supplied, the expected_delta should be the deleted sequence. If expected_delta is not NULL, the function checks this and aborts if this is not the case.

Dually, pos is the 1-based position in short string in front of which one can make an insertion to get long_str.

This function moves pos as far to left as possible so that a deletion at that position still results in an edit of long_str to generate short_str.

Usage

```
justify_indel(long_str, short_str, pos, expected_delta = NULL)
```

Arguments

long_str	A single character string
short_str	A single character string.
pos	An integer; see the description
expected_delta	A single string; see the description

Value

A list with elements

- leftmost_pos
- del_string
- del_string_plus del_string plus one base to the left the position of that base to the left is leftmost_pos - 1
- edge_warning See the third example. We are not sure how far to the left we could have moved the indel if we had more context on the left.
- error if non-NULL we did not see the expected value for expected_delta

Examples

```
justify_indel("CAAAG", "CAAG", pos = 2, expected_delta = "A")
justify_indel("CAAAG", "CAAG", pos = 3, expected_delta = "A")
justify_indel("CACAG", "CAG", pos = 3, expected_delta = "CA")
```

MutectVCFFilesToCatalog

Create SBS, DBS and Indel catalogs from Mutect VCF files

Description

Deprecated, use VCFsToCatalogs(variant.caller = "mutect") instead Create 3 SBS catalogs (96, 192, 1536), 3 DBS catalogs (78, 136, 144) and Indel catalog from the Mutect VCFs specified by files

Usage

```
MutectVCFFilesToCatalog(
  files,
  ref.genome,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  tumor.col.names = NA,
  flag.mismatches = 0,
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

<code>files</code>	Character vector of file paths to the Mutect VCF files.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>trans.ranges</code>	Optional. If <code>ref.genome</code> specifies one of the BSgenome object 1. <code>BSgenome.Hsapiens.1000genomes.hs37d5</code> 2. <code>BSgenome.Hsapiens.UCSC.hg38</code> 3. <code>BSgenome.Mmusculus.UCSC.mm10</code> then the function will infer <code>trans.ranges</code> automatically. Otherwise, user will need to provide the necessary <code>trans.ranges</code> . Please refer to TranscriptRanges for more details. If <code>is.null(trans.ranges)</code> do not add transcript range information.
<code>region</code>	A character string designating a genomic region; see as.catalog and ICAMS .
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCF file paths in <code>files</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>files</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
<code>tumor.col.names</code>	Optional. Vector of column names or column indices in VCFs which contain the tumor sample information. The order of elements in <code>tumor.col.names</code> should match the order of VCFs specified in <code>files</code> . If <code>tumor.col.names</code> is equal to <code>NA</code> (default), this function will use the 10th column in all the VCFs to calculate VAFs. See GetMutectVAF for more details.
<code>flag.mismatches</code>	Deprecated. If there are ID variants whose REF do not match the extracted sequence from <code>ref.genome</code> , the function will automatically discard these variants and an element <code>discarded.variants</code> will appear in the return value. See AnnotateIDVCF for more details.
<code>return.annotated.vcfs</code>	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is <code>FALSE</code> .
<code>suppress.discarded.variants.warnings</code>	Logical. Whether to suppress warning messages showing information about the discarded variants. Default is <code>TRUE</code> .

Details

This function calls [VCFsToSBSCatalogs](#), [VCFsToDBSCatalogs](#) and [VCFsToIDCatalogs](#)

Value

A list containing the following objects:

- `catSBS96`, `catSBS192`, `catSBS1536`: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- `catDBS78`, `catDBS136`, `catDBS144`: Matrix of 3 DBS catalogs (one each for 78, 136, and 144).
- `catID`: Matrix of ID (small insertions and deletions) catalog.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of elements:
 - SBS: SBS VCF annotated by [AnnotateSBSCat](#) with three new columns `SBS96.class`, `SBS192.class` and `SBS1536.class` showing the mutation class for each SBS variant.
 - DBS: DBS VCF annotated by [AnnotateDBSCat](#) with three new columns `DBS78.class`, `DBS136.class` and `DBS144.class` showing the mutation class for each DBS variant.
 - ID: ID VCF annotated by [AnnotateIDCat](#) with one new column `ID.class` showing the mutation class for each ID variant.

If `trans.ranges` is not provided by user and cannot be inferred by ICAMS, SBS 192 and DBS 144 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

SBS 192 and DBS 144 catalogs include only mutations in transcribed regions. In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Comments

To add or change attributes of the catalog, you can use function [attr](#). For example, `attr(catalog, "abundance") <- custom.abundance`.

Examples

```
## Not run:
file <- c(system.file("extdata/Mutect-vcf",
                      "Mutect.GRCh37.s1.vcf",
                      package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
```

```
catalogs <- MutectVCFFilesToCatalog(file, ref.genome = "hg19",
                                   trans.ranges = trans.ranges.GRCh37,
                                   region = "genome")

## End(Not run)
```

MutectVCFFilesToCatalogAndPlotToPdf

[Deprecated, use VCFsToCatalogsAndPlotToPdf(variant.caller = "mutect") instead] Create SBS, DBS and Indel catalogs from Mutect VCF files and plot them to PDF

Description

[Deprecated, use VCFsToCatalogsAndPlotToPdf(variant.caller = "mutect") instead] Create 3 SBS catalogs (96, 192, 1536), 3 DBS catalogs (78, 136, 144) and Indel catalog from the Mutect VCFs specified by files and plot them to PDF

Usage

```
MutectVCFFilesToCatalogAndPlotToPdf(
  files,
  ref.genome,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  tumor.col.names = NA,
  output.file = "",
  flag.mismatches = 0,
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

files	Character vector of file paths to the Mutect VCF files.
ref.genome	A ref.genome argument as described in ICAMS .
trans.ranges	Optional. If ref.genome specifies one of the BSgenome object 1. BSgenome.Hsapiens.1000genomes.hs37d5 2. BSgenome.Hsapiens.UCSC.hg38 3. BSgenome.Mmusculus.UCSC.mm10 then the function will infer trans.ranges automatically. Otherwise, user will need to provide the necessary trans.ranges. Please refer to TranscriptRanges for more details. If is.null(trans.ranges) do not add transcript range information.
region	A character string designating a genomic region; see as.catalog and ICAMS .
names.of.VCFs	Optional. Character vector of names of the VCF files. The order of names in names.of.VCFs should match the order of VCF file paths in files. If NULL(default), this function will remove all of the path up to and including the last path separator (if any) in files and file paths without extensions (and the leading dot) will be used as the names of the VCF files.

tumor.col.names	Optional. Vector of column names or column indices in VCFs which contain the tumor sample information. The order of elements in tumor.col.names should match the order of VCFs specified in files. If tumor.col.names is equal to NA(default), this function will use the 10th column in all the VCFs to calculate VAFs. See GetMutectVAF for more details.
output.file	Optional. The base name of the PDF files to be produced; multiple files will be generated, each ending in <i>x.pdf</i> , where <i>x</i> indicates the type of catalog plotted in the file.
flag.mismatches	Deprecated. If there are ID variants whose REF do not match the extracted sequence from ref.genome, the function will automatically discard these variants and an element discarded.variants will appear in the return value. See AnnotateIDVCF for more details.
return.annotated.vcfs	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.
suppress.discarded.variants.warnings	Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

Details

This function calls [MutectVCFFilesToCatalog](#) and [PlotCatalogToPdf](#)

Value

A list containing the following objects:

- catSBS96, catSBS192, catSBS1536: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- catDBS78, catDBS136, catDBS144: Matrix of 3 DBS catalogs (one each for 78, 136, and 144).
- catID: Matrix of ID (small insertions and deletions) catalog.
- discarded.variants: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column discarded.reason for more details.
- annotated.vcfs: **Non-NULL only if** return.annotated.vcfs = TRUE. A list of elements:
 - SBS: SBS VCF annotated by [AnnotateSBSVCF](#) with three new columns SBS96.class, SBS192.class and SBS1536.class showing the mutation class for each SBS variant.
 - DBS: DBS VCF annotated by [AnnotateDBSVCF](#) with three new columns DBS78.class, DBS136.class and DBS144.class showing the mutation class for each DBS variant.
 - ID: ID VCF annotated by [AnnotateIDVCF](#) with one new column ID.class showing the mutation class for each ID variant.

If trans.ranges is not provided by user and cannot be inferred by ICAMS, SBS 192 and DBS 144 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

Note

SBS 192 and DBS 144 catalogs include only mutations in transcribed regions. In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Comments

To add or change attributes of the catalog, you can use function `attr`.
For example, `attr(catalog, "abundance") <- custom.abundance`.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for `Canonicalize1Del` which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see `FindMaxRepeatDel`), and if the deletion is not in a simple repeat, looks for microhomology (see `FindDelMH`).

Examples

```
## Not run:
file <- c(system.file("extdata/Mutect-vcf",
                     "Mutect.GRCh37.s1.vcf",
                     package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs <-
    MutectVCFFilesToCatalogAndPlotToPdf(file, ref.genome = "hg19",
                                       trans.ranges = trans.ranges.GRCh37,
                                       region = "genome",
                                       output.file =
                                         file.path(tempdir(), "Mutect"))}

## End(Not run)
```

MutectVCFFilesToZipFile

Create a zip file which contains catalogs and plot PDFs from Mutect VCF files

Description

Deprecated, use `VCFsToZipFile(variant.caller = "mutect")` instead Create 3 SBS catalogs (96, 192, 1536), 3 DBS catalogs (78, 136, 144) and Indel catalog from the Mutect VCFs specified by `dir`, save the catalogs as CSV files, plot them to PDF and generate a zip archive of all the output files.

Usage

```
MutectVCFFilesToZipFile(
  dir,
  zipfile,
  ref.genome,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  tumor.col.names = NA,
```

```

base.filename = "",
flag.mismatches = 0,
return.annotated.vcfs = FALSE,
suppress.discarded.variants.warnings = TRUE
)

```

Arguments

<code>dir</code>	Pathname of the directory which contains only the Mutect VCF files. Each Mutect VCF must have a file extension ".vcf" (case insensitive) and share the same <code>ref.genome</code> and region.
<code>zipfile</code>	Pathname of the zip file to be created.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>trans.ranges</code>	Optional. If <code>ref.genome</code> specifies one of the BSgenome object 1. <code>BSgenome.Hsapiens.1000genomes.hs37d5</code> 2. <code>BSgenome.Hsapiens.UCSC.hg38</code> 3. <code>BSgenome.Mmusculus.UCSC.mm10</code> then the function will infer <code>trans.ranges</code> automatically. Otherwise, user will need to provide the necessary <code>trans.ranges</code> . Please refer to TranscriptRanges for more details. If <code>is.null(trans.ranges)</code> do not add transcript range information.
<code>region</code>	A character string designating a genomic region; see as.catalog and ICAMS .
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCFs listed in <code>dir</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>dir</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
<code>tumor.col.names</code>	Optional. Vector of column names or column indices in VCFs which contain the tumor sample information. The order of elements in <code>tumor.col.names</code> should match the order of VCFs listed in <code>dir</code> . If <code>tumor.col.names</code> is equal to <code>NA</code> (default), this function will use the 10th column in all the VCFs to calculate VAFs. See GetMutectVAF for more details.
<code>base.filename</code>	Optional. The base name of the CSV and PDF files to be produced; multiple files will be generated, each ending in <code>x.csv</code> or <code>x.pdf</code> , where <code>x</code> indicates the type of catalog.
<code>flag.mismatches</code>	Deprecated. If there are ID variants whose REF do not match the extracted sequence from <code>ref.genome</code> , the function will automatically discard these variants and an element <code>discarded.variants</code> will appear in the return value. See AnnotateIDVCF for more details.
<code>return.annotated.vcfs</code>	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is <code>FALSE</code> .
<code>suppress.discarded.variants.warnings</code>	Logical. Whether to suppress warning messages showing information about the discarded variants. Default is <code>TRUE</code> .

Details

This function calls [MutectVCFFilesToCatalog](#), [PlotCatalogToPdf](#), [WriteCatalog](#) and `zip::zipr`.

Value

A list containing the following objects:

- catSBS96, catSBS192, catSBS1536: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- catDBS78, catDBS136, catDBS144: Matrix of 3 DBS catalogs (one each for 78, 136, and 144).
- catID: Matrix of ID (small insertions and deletions) catalog.
- discarded.variants: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column discarded.reason for more details.
- annotated.vcfs: **Non-NULL only if** return.annotated.vcfs = TRUE. A list of elements:
 - SBS: SBS VCF annotated by [AnnotateSBSVCF](#) with three new columns SBS96.class, SBS192.class and SBS1536.class showing the mutation class for each SBS variant.
 - DBS: DBS VCF annotated by [AnnotateDBSVCF](#) with three new columns DBS78.class, DBS136.class and DBS144.class showing the mutation class for each DBS variant.
 - ID: ID VCF annotated by [AnnotateIDVCF](#) with one new column ID.class showing the mutation class for each ID variant.

If trans.ranges is not provided by user and cannot be inferred by ICAMS, SBS 192 and DBS 144 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

SBS 192 and DBS 144 catalogs include only mutations in transcribed regions. In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Comments

To add or change attributes of the catalog, you can use function [attr](#).
For example, attr(catalog, "abundance") <- custom.abundance.

Examples

```
## Not run:
dir <- c(system.file("extdata/Mutect-vcf",
  package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs <-
    MutectVCFFilesToZipFile(dir,
      zipfile = file.path(tempdir(), "test.zip"),
      ref.genome = "hg19",
      trans.ranges = trans.ranges.GRCh37,
```

```

        region = "genome",
        base.filename = "Mutect")
unlink(file.path(tempdir(), "test.zip"))}

## End(Not run)

```

PlotCatalog

*Plot **one** spectrum or signature*

Description

Plot the spectrum of **one** sample or plot **one** signature. The type of graph is based on `attribute("catalog.type")` of the input catalog. You can first use [TransformCatalog](#) to get different types of catalog and then do the plotting.

Usage

```

PlotCatalog(
  catalog,
  plot.SBS12 = NULL,
  cex = NULL,
  grid = NULL,
  upper = NULL,
  xlabels = NULL,
  ylabels = NULL,
  ylim = NULL
)

```

Arguments

catalog	A catalog as defined in ICAMS with attributes added. See as.catalog for more details. catalog can also be a numeric matrix, numeric data.frame, or a vector denoting the mutation counts , but must be in the correct row order used in ICAMS . See CatalogRowOrder for more details. If catalog is a vector, it will be converted to a 1-column matrix with rownames taken from the element names of the vector and with column name "Unknown".
plot.SBS12	Only meaningful for class SBS192Catalog; if TRUE, generate an abbreviated plot of only SBS without context, i.e. C>A, C>G, C>T, T>A, T>C, T>G each on transcribed and untranscribed strands, rather than SBS in trinucleotide context, e.g. ACA > AAA, ACA > AGA, ..., TCT > TAT, ... There are 12 bars in the graph.
cex	Has the usual meaning. Taken from <code>par("cex")</code> by default. Only implemented for SBS96Catalog, SBS192Catalog and DBS144Catalog.
grid	A logical value indicating whether to draw grid lines. Only implemented for SBS96Catalog, DBS78Catalog, IndelCatalog, ID166Catalog.
upper	A logical value indicating whether to draw horizontal lines and the names of major mutation class on top of graph. Only implemented for SBS96Catalog, DBS78Catalog, IndelCatalog, ID166Catalog.
xlabels	A logical value indicating whether to draw x axis labels. Only implemented for SBS96Catalog, DBS78Catalog, IndelCatalog, ID166Catalog. If FALSE then plot x axis tick marks for SBS96Catalog; set <code>par(tck = 0)</code> to suppress.

ylabels	A logical value indicating whether to draw y axis labels. Only implemented for SBS96Catalog, DBS78Catalog, IndelCatalog, ID166Catalog.
ylim	Has the usual meaning. Only implemented for SBS96Catalog, IndelCatalog, ID166Catalog.

Value

An **invisible** list whose first element is a logic value indicating whether the plot is successful. For SBS96Catalog, SBS192Catalog, DBS78Catalog, DBS144Catalog and IndelCatalog, the list will have a second element, which is a numeric vector giving the coordinates of all the bar midpoints drawn, useful for adding to the graph. For **SBS192Catalog** with "counts" catalog.type and non-NULL abundance and plot.SBS12 = TRUE, the list will have an additional element which is a list containing the strand bias statistics.

Comments

For **SBS192Catalog** with "counts" catalog.type and non-NULL abundance and plot.SBS12 = TRUE, the strand bias statistics are Benjamini-Hochberg q-values based on two-sided binomial tests of the mutation counts on the transcribed and untranscribed strands relative to the actual abundances of C and T on the transcribed strand. On the SBS12 plot, asterisks indicate q-values as follows *, $Q < 0.05$; **, $Q < 0.01$; ***, $Q < 0.001$.

Note

The sizes of repeats involved in deletions range from 0 to 5+ in the mutational-spectra and signature catalog rownames, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Examples

```
file <- system.file("extdata",
                    "strelka.regress.cat.sbs.96.csv",
                    package = "ICAMS")
catSBS96 <- ReadCatalog(file)
colnames(catSBS96) <- "sample"
PlotCatalog(catSBS96)
```

PlotCatalogToPdf

Plot catalog to a PDF file

Description

Plot catalog to a PDF file. The type of graph is based on attribute("catalog.type") of the input catalog. You can first use [TransformCatalog](#) to get different types of catalog and then do the plotting.

Usage

```
PlotCatalogToPdf(
  catalog,
  file,
  plot.SBS12 = NULL,
  cex = NULL,
  grid = NULL,
  upper = NULL,
  xlabels = NULL,
  ylabels = NULL,
  ylim = NULL
)
```

Arguments

catalog	A catalog as defined in ICAMS with attributes added. See as.catalog for more details. catalog can also be a numeric matrix, numeric data.frame, or a vector denoting the mutation counts , but must be in the correct row order used in ICAMS . See CatalogRowOrder for more details. If catalog is a vector, it will be converted to a 1-column matrix with rownames taken from the element names of the vector and with column name "Unknown".
file	The name of the PDF file to be produced.
plot.SBS12	Only meaningful for class SBS192Catalog; if TRUE, generate an abbreviated plot of only SBS without context, i.e. C>A, C>G, C>T, T>A, T>C, T>G each on transcribed and untranscribed strands, rather than SBS in trinucleotide context, e.g. ACA > AAA, ACA > AGA, ..., TCT > TAT, ... There are 12 bars in the graph.
cex	Has the usual meaning. Taken from par("cex") by default. Only implemented for SBS96Catalog, SBS192Catalog and DBS144Catalog.
grid	A logical value indicating whether to draw grid lines. Only implemented for SBS96Catalog, DBS78Catalog, IndelCatalog, ID166Catalog.
upper	A logical value indicating whether to draw horizontal lines and the names of major mutation class on top of graph. Only implemented for SBS96Catalog, DBS78Catalog, IndelCatalog, ID166Catalog.
xlabels	A logical value indicating whether to draw x axis labels. Only implemented for SBS96Catalog, DBS78Catalog, IndelCatalog, ID166Catalog. If FALSE then plot x axis tick marks for SBS96Catalog; set par(tck = 0) to suppress.
ylabels	A logical value indicating whether to draw y axis labels. Only implemented for SBS96Catalog, DBS78Catalog, IndelCatalog, ID166Catalog.
ylim	Has the usual meaning. Only implemented for SBS96Catalog, IndelCatalog, ID166Catalog.

Value

An **invisible** list whose first element is a logic value indicating whether the plot is successful. For **SBS192Catalog** with "counts" catalog.type and non-null abundance and plot.SBS12 = TRUE, the list will have a second element which is a list containing the strand bias statistics.

Comments

For **SBS192Catalog** with "counts" catalog.type and non-NULL abundance and plot.SBS12 = TRUE, the strand bias statistics are Benjamini-Hochberg q-values based on two-sided binomial tests of the mutation counts on the transcribed and untranscribed strands relative to the actual abundances of C and T on the transcribed strand. On the SBS12 plot, asterisks indicate q-values as follows *, $Q < 0.05$; **, $Q < 0.01$; ***, $Q < 0.001$.

Note

The sizes of repeats involved in deletions range from 0 to 5+ in the mutational-spectra and signature catalog rownames, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Examples

```
file <- system.file("extdata",
                    "strelka.regress.cat.sbs.96.csv",
                    package = "ICAMS")
catSBS96 <- ReadCatalog(file)
colnames(catSBS96) <- "sample"
PlotCatalogToPdf(catSBS96, file = file.path(tempdir(), "test.pdf"))
```

PlotTransBiasGeneExp *Plot transcription strand bias with respect to gene expression values*

Description

Plot transcription strand bias with respect to gene expression values

Usage

```
PlotTransBiasGeneExp(
  annotated.SBS.vcf,
  expression.data,
  Ensembl.gene.ID.col,
  expression.value.col,
  num.of.bins,
  plot.type,
  damaged.base = NULL,
  ymax = NULL
)
```

Arguments

annotated.SBS.vcf

An SBS VCF annotated by [AnnotateSBSVCF](#). It **must** have transcript range information added.

expression.data

A [data.table](#) which contains the expression values of genes.
See [GeneExpressionData](#) for more details.

Ensembl.gene.ID.col

Name of column which has the Ensembl gene ID information in expression.data.

expression.value.col	Name of column which has the gene expression values in expression.data.
num.of.bins	The number of bins that will be plotted on the graph.
plot.type	A character string indicating one mutation type to be plotted. It should be one of "C>A", "C>G", "C>T", "T>A", "T>C", "T>G".
damaged.base	One of NULL, "purine" or "pyrimidine". This function allocates approximately equal numbers of mutations from damaged.base into each of num.of.bins bin by expression level. E.g. if damaged.base is "purine", then mutations from A and G will be allocated in approximately equal numbers to each expression-level bin. The rationale for the name damaged.base is that the direction of strand bias is a result of whether the damage occurs on a purine or pyrimidine. If NULL, the function attempts to infer the damaged.base based on mutation counts.
ymax	Limit for the y axis. If not specified, it defaults to NULL and the y axis limit equals 1.5 times of the maximum mutation counts in a specific mutation type.

Value

A list whose first element is a logic value indicating whether the plot is successful. The second element is a named numeric vector containing the p-values printed on the plot.

Note

The p-values are calculated by logistic regression using function `glm`. The dependent variable is labeled "1" and "0" if the mutation from annotated.SBS.vcf falls onto the untranscribed and transcribed strand respectively. The independent variable is the binary logarithm of the gene expression value from expression.data plus one, i.e. $\log_2(x + 1)$ where x stands for gene expression value.

Examples

```
file <- c(system.file("extdata/Strelka-SBS-vcf/",
                     "Strelka.SBS.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadAndSplitVCFs(file, variant.caller = "strelka")
SBS.vcf <- list.of.vcfs$SBS[[1]]
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  annotated.SBS.vcf <- AnnotateSBSVCF(SBS.vcf, ref.genome = "hg19",
                                     trans.ranges = trans.ranges.GRCh37)
  PlotTransBiasGeneExp(annotated.SBS.vcf = annotated.SBS.vcf,
                      expression.data = gene.expression.data.HepG2,
                      Ensembl.gene.ID.col = "Ensembl.gene.ID",
                      expression.value.col = "TPM",
                      num.of.bins = 4, plot.type = "C>A")
}
```

PlotTransBiasGeneExpToPdf

*Plot transcription strand bias with respect to gene expression values
to a PDF file*

Description

Plot transcription strand bias with respect to gene expression values to a PDF file

Usage

```
PlotTransBiasGeneExpToPdf(
  annotated.SBS.vcf,
  file,
  expression.data,
  Ensembl.gene.ID.col,
  expression.value.col,
  num.of.bins,
  plot.type = c("C>A", "C>G", "C>T", "T>A", "T>C", "T>G"),
  damaged.base = NULL
)
```

Arguments

<code>annotated.SBS.vcf</code>	An SBS VCF annotated by AnnotateSBSVCF . It must have transcript range information added.
<code>file</code>	The name of output file.
<code>expression.data</code>	A data.table which contains the expression values of genes. See GeneExpressionData for more details.
<code>Ensembl.gene.ID.col</code>	Name of column which has the Ensembl gene ID information in <code>expression.data</code> .
<code>expression.value.col</code>	Name of column which has the gene expression values in <code>expression.data</code> .
<code>num.of.bins</code>	The number of bins that will be plotted on the graph.
<code>plot.type</code>	A vector of character indicating types to be plotted. It can be one or more types from "C>A", "C>G", "C>T", "T>A", "T>C", "T>G". The default is to print all the six mutation types.
<code>damaged.base</code>	One of NULL, "purine" or "pyrimidine". This function allocates approximately equal numbers of mutations from <code>damaged.base</code> into each of <code>num.of.bins</code> bin by expression level. E.g. if <code>damaged.base</code> is "purine", then mutations from A and G will be allocated in approximately equal numbers to each expression-level bin. The rationale for the name <code>damaged.base</code> is that the direction of strand bias is a result of whether the damage occurs on a purine or pyrimidine. If NULL, the function attempts to infer the <code>damaged.base</code> based on mutation counts.

Value

A list whose first element is a logic value indicating whether the plot is successful. The second element is a named numeric vector containing the p-values printed on the plot.

Note

The p-values are calculated by logistic regression using function [glm](#). The dependent variable is labeled "1" and "0" if the mutation from `annotated.SBS.vcf` falls onto the untranscribed and tran-

scribed strand respectively. The independent variable is the binary logarithm of the gene expression value from expression.data plus one, i.e. $\log_2(x + 1)$ where x stands for gene expression value.

Examples

```
file <- c(system.file("extdata/Strelka-SBS-vcf/",
                     "Strelka.SBS.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadAndSplitVCFs(file, variant.caller = "strelka")
SBS.vcf <- list.of.vcfs$SBS[[1]]
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  annotated.SBS.vcf <- AnnotateSBSVCF(SBS.vcf, ref.genome = "hg19",
                                     trans.ranges = trans.ranges.GRCh37)
  PlotTransBiasGeneExpToPdf(annotated.SBS.vcf = annotated.SBS.vcf,
                            expression.data = gene.expression.data.HepG2,
                            Ensembl.gene.ID.col = "Ensembl.gene.ID",
                            expression.value.col = "TPM",
                            num.of.bins = 4,
                            plot.type = c("C>A", "C>G", "C>T", "T>A", "T>C"),
                            file = file.path(tempdir(), "test.pdf"))
}
```

ReadAndSplitMutectVCFs

[Deprecated, use ReadAndSplitVCFs(variant.caller = "mutect") instead] *Read and split Mutect VCF files*

Description

[Deprecated, use ReadAndSplitVCFs(variant.caller = "mutect") instead] Read and split Mutect VCF files

Usage

```
ReadAndSplitMutectVCFs(
  files,
  names.of.VCFs = NULL,
  tumor.col.names = NA,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

<code>files</code>	Character vector of file paths to the Mutect VCF files.
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCF file paths in <code>files</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>files</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
<code>tumor.col.names</code>	Optional. Vector of column names or column indices in VCFs which contain the tumor sample information. The order of elements in <code>tumor.col.names</code> should

match the order of VCFs specified in files. If `tumor.col.names` is equal to `NA`(default), this function will use the 10th column in all the VCFs to calculate VAFs. See [GetMutectVAF](#) for more details.

`suppress.discarded.variants.warnings`

Logical. Whether to suppress warning messages showing information about the discarded variants. Default is `TRUE`.

Value

A list containing the following objects:

- `SBS`: List of VCFs with only single base substitutions.
- `DBS`: List of VCFs with only doublet base substitutions as called by Mutect.
- `ID`: List of VCFs with only small insertions and deletions.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.

See Also

[MutectVCFFilesToCatalog](#)

Examples

```
## Not run:
file <- c(system.file("extdata/Mutect-vcf",
                     "Mutect.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadAndSplitMutectVCFs(file)

## End(Not run)
```

ReadAndSplitStrelkaSBSVCFs

[Deprecated, use `ReadAndSplitVCFs(variant.caller = "strelka")` instead] *Read and split Strelka SBS VCF files*

Description

[Deprecated, use `ReadAndSplitVCFs(variant.caller = "strelka")` instead] The function will find and merge adjacent SBS pairs into DBS if their VAFs are very similar. The default threshold value for VAF is 0.02.

Usage

```
ReadAndSplitStrelkaSBSVCFs(
  files,
  names.of.VCFs = NULL,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

- `files` Character vector of file paths to the Strelka SBS VCF files.
- `names.of.VCFs` Optional. Character vector of names of the VCF files. The order of names in `names.of.VCFs` should match the order of VCF file paths in `files`. If `NULL` (default), this function will remove all of the path up to and including the last path separator (if any) in `files` and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
- `suppress.discarded.variants.warnings` Logical. Whether to suppress warning messages showing information about the discarded variants. Default is `TRUE`.

Value

A list of elements as follows:

- `SBS.vcfs`: List of data.frames of pure SBS mutations – no DBS or 3+BS mutations.
- `DBS.vcfs`: List of data.frames of pure DBS mutations – no SBS or 3+BS mutations.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.

See Also

[StrelkaSBSVCFFilesToCatalog](#)

Examples

```
## Not run:
file <- c(system.file("extdata/Strelka-SBS-vcf",
                     "Strelka.SBS.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadAndSplitStrelkaSBSVCFs(file)

## End(Not run)
```

ReadAndSplitVCFs	<i>Read and split VCF files</i>
------------------	---------------------------------

Description

Read and split VCF files

Usage

```
ReadAndSplitVCFs(
  files,
  variant.caller = "unknown",
  num.of.cores = 1,
  names.of.VCFs = NULL,
  tumor.col.names = NA,
  filter.status = DefaultFilterStatus(variant.caller),
  get.vaf.function = NULL,
```



```

    ...,
    max.vaf.diff = 0.02,
    suppress.discarded.variants.warnings = TRUE,
    always.merge.SBS = FALSE,
    chr.names.to.process = NULL
)

```

Arguments

- | | |
|------------------|---|
| files | Character vector of file paths to the VCF files. |
| variant.caller | Name of the variant caller that produces the VCF, can be either "strelka", "mutect", "freebayes" or "unknown". This information is needed to calculate the VAFs (variant allele frequencies). If variant caller is "unknown" (default) and get.vaf.function is NULL, then VAF and read depth will be NAs. If variant caller is "mutect", do not merge SBSs into DBS. |
| num.of.cores | The number of cores to use. Not available on Windows unless num.of.cores = 1. |
| names.of.VCFs | Optional. Character vector of names of the VCF files. The order of names in names.of.VCFs should match the order of VCF file paths in files. If NULL (default), this function will remove all of the path up to and including the last path separator (if any) in files and file paths without extensions (and the leading dot) will be used as the names of the VCF files. |
| tumor.col.names | Optional. Only applicable to Mutect VCFs. Vector of column names or column indices in Mutect VCFs which contain the tumor sample information. The order of elements in tumor.col.names should match the order of Mutect VCFs specified in files. If tumor.col.names is equal to NA (default), this function will use the 10th column in all the Mutect VCFs to calculate VAFs. See GetMutectVAF for more details. |
| filter.status | The character string in column FILTER of the VCF that indicates that a variant has passed all the variant caller's filters. Variants (lines in the VCF) for which the value in column FILTER does not equal filter.status are silently excluded from the output. The internal function DefaultFilterStatus tries to infer filter.status based on variant.caller. If variant.caller is "unknown", user must specify filter.status explicitly. If filter.status = NULL, all variants are retained. If there is no FILTER column in the VCF, all variants are retained with a warning. |
| get.vaf.function | Optional. Only applicable when variant.caller is " unknown ". Function to calculate VAF (variant allele frequency) and read depth information from original VCF. See GetMutectVAF as an example. If NULL (default) and variant.caller is "unknown", then VAF and read depth will be NAs. |
| ... | Optional arguments to get.vaf.function. |
| max.vaf.diff | Not applicable if variant.caller = "mutect". The maximum difference of VAF, default value is 0.02. If the absolute difference of VAFs for adjacent SBSs is bigger than max.vaf.diff, then these adjacent SBSs are likely to be "merely" asynchronous single base mutations, opposed to a simultaneous doublet mutation or variants involving more than two consecutive bases. Use negative value (e.g. -1) to suppress merging adjacent SBSs to DBS. |

`suppress.discarded.variants.warnings`
 Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

`always.merge.SBS`
 If TRUE merge adjacent SBSs as DBSs regardless of VAFs and regardless of the value of `max.vaf.diff` and regardless of the value of `get.vaf.function`. It is an error to set this to TRUE when `variant.caller = "mutect"`.

`chr.names.to.process`
 A character vector specifying the chromosome names in VCF whose variants will be kept and processed, other chromosome variants will be discarded. If NULL(default), all variants will be kept except those on chromosomes with names that contain strings "GL", "KI", "random", "Hs", "M", "JH", "fix", "alt".

Value

A list containing the following objects:

- SBS: List of VCFs with only single base substitutions.
- DBS: List of VCFs with only doublet base substitutions.
- ID: List of VCFs with only small insertions and deletions.
- discarded.variants: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.

See Also

[VCFsToCatalogs](#)

Examples

```
file <- c(system.file("extdata/Mutect-vcf",
                     "Mutect.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadAndSplitVCFs(file, variant.caller = "mutect")
```

ReadCatalog

Read catalog

Description

Read a catalog in standardized format from path.

Usage

```
ReadCatalog(
  file,
  ref.genome = NULL,
  region = "unknown",
  catalog.type = "counts",
  strict = NULL,
  stop.on.error = TRUE
)
```

Arguments

<code>file</code>	Path to a catalog on disk in a standardized format. The recognized formats are: • ICAMS formatted SBS96, SBS192, SBS1536, DBS78, DBS136, DBS144, ID, ID166 (see CatalogRowOrder). • SigProfiler-formatted SBS96, DBS78 and ID83 catalogs; see https://github.com/AlexandrovLab/SigProfilerExtractor. • COSMIC-formatted SBS96, SBS192 (a.k.a. TSB192), DBS78, ID83 catalogs; see https://cancer.sanger.ac.uk/signatures/.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>region</code>	region A character string designating a genomic region; see as.catalog and ICAMS .
<code>catalog.type</code>	One of "counts", "density", "counts.signature", "density.signature".
<code>strict</code>	Ignored and deprecated.
<code>stop.on.error</code>	If TRUE, call stop on error; otherwise return a 1-column matrix of NA's with the attribute "error" containing error information. The number of rows may not be the correct number for the expected catalog type.

Details

See also [WriteCatalog](#)

Value

A catalog as an S3 object; see [as.catalog](#).

Comments

To add or change attributes of the catalog, you can use function [attr](#).
For example, `attr(catalog, "abundance") <- custom.abundance`.

Note

In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Examples

```
file <- system.file("extdata",
                    "strelka.regress.cat.sbs.96.csv",
                    package = "ICAMS")
catSBS96 <- ReadCatalog(file)
```

ReadStrelkaIDVCFs	<i>Read Strelka ID (small insertions and deletions) VCF files</i>
-------------------	---

Description

Deprecated, use `ReadAndSplitVCFs(variant.caller = "strelka")` instead

Usage

```
ReadStrelkaIDVCFs(files, names.of.VCFs = NULL)
```

Arguments

<code>files</code>	Character vector of file paths to the VCF files.
<code>names.of.VCFs</code>	Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCF file paths in <code>files</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>files</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.

Value

A list of data frames containing data lines of the VCF files.

Note

In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

See Also

[StrelkaIDVCFFilesToCatalog](#)

Examples

```
## Not run:
file <- c(system.file("extdata/Strelka-ID-vcf",
                     "Strelka.ID.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadStrelkaIDVCFs(file)

## End(Not run)
```

ReadVCFs

*Read VCF files***Description**

Read VCF files

Usage

```
ReadVCFs(
  files,
  variant.caller = "unknown",
  num.of.cores = 1,
  names.of.VCFs = NULL,
  tumor.col.names = NA,
  filter.status = DefaultFilterStatus(variant.caller),
  get.vaf.function = NULL,
  ...
)
```

Arguments

<code>files</code>	Character vector of file paths to the VCF files.
<code>variant.caller</code>	Name of the variant caller that produces the VCF, can be either "strelka", "mutect", "freebayes" or "unknown". This information is needed to calculate the VAFs (variant allele frequencies). If variant caller is "unknown" (default) and <code>get.vaf.function</code> is NULL, then VAF and read depth will be NAs. If variant caller is "mutect", do not merge SBSs into DBS.
<code>num.of.cores</code>	The number of cores to use. Not available on Windows unless <code>num.of.cores = 1</code> .
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCF file paths in <code>files</code> . If NULL (default), this function will remove all of the path up to and including the last path separator (if any) in <code>files</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
<code>tumor.col.names</code>	Optional. Only applicable to Mutect VCFs. Vector of column names or column indices in Mutect VCFs which contain the tumor sample information. The order of elements in <code>tumor.col.names</code> should match the order of Mutect VCFs specified in <code>files</code> . If <code>tumor.col.names</code> is equal to NA (default), this function will use the 10th column in all the Mutect VCFs to calculate VAFs. See GetMutectVAF for more details.
<code>filter.status</code>	The character string in column FILTER of the VCF that indicates that a variant has passed all the variant caller's filters. Variants (lines in the VCF) for which the value in column FILTER does not equal <code>filter.status</code> are silently excluded from the output. The internal function <code>DefaultFilterStatus</code> tries to infer <code>filter.status</code> based on <code>variant.caller</code> . If <code>variant.caller</code> is "unknown", user must specify <code>filter.status</code> explicitly. If <code>filter.status = NULL</code> , all variants are retained. If there is no FILTER column in the VCF, all variants are retained with a warning.

`get.vaf.function`

Optional. Only applicable when `variant.caller` is **"unknown"**. Function to calculate VAF(variant allele frequency) and read depth information from original VCF. See [GetMutectVAF](#) as an example. If `NULL`(default) and `variant.caller` is "unknown", then VAF and read depth will be NAs.

...

Optional arguments to `get.vaf.function`.

Value

A list of data frames storing data lines of the VCF files with two additional columns added which contain the VAF(variant allele frequency) and read depth information.

Examples

```
file <- c(system.file("extdata/Mutect-vcf",
                     "Mutect.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadVCFs(file, variant.caller = "mutect")
```

revc

Reverse complement every string in string.vec

Description

Based on [reverseComplement](#). Handles IUPAC ambiguity codes but not "u" (uracil). (see https://en.wikipedia.org/wiki/Nucleic_acid_notation).

Usage

```
revc(string.vec)
```

Arguments

`string.vec` A character vector.

Value

A character vector with the reverse complement of every string in `string.vec`.

Examples

```
revc("aTgc") # GCAT

# A vector and strings with ambiguity codes
revc(c("ATGC", "aTgc", "wnTCb")) # GCAT GCAT VGANW

## Not run:
revc("ACGU") # An error
## End(Not run)
```

segment_simple_cpp	<i>Segment a single indel using Rcpp interface</i>
--------------------	--

Description

This function provides direct C++ interface for indel segmentation without calling an external binary via system2.

Usage

```
segment_simple_cpp(ins_or_del, string, context)
```

Arguments

ins_or_del	Character indicating insertion ("i") or deletion ("d")
string	A single indel sequence to segment (character scalar)
context	A single flanking context sequence (character scalar)

Value

A list with the following elements:

unit	Repeat unit sequence (character)
unit_length	Unit length (integer)
internal_rep	Internal repeat region (character)
internal_reps	Internal repeat count (integer)
spacer	Spacer sequence (character)
spacer_length	Spacer length (integer)
prime3_rep	3' flanking repeat region (character)
prime3_reps	3' flanking repeat count (integer)
original_reps	Original repeat count (integer)

seg_simple	<i>Segment a single indel sequence using Rcpp interface</i>
------------	---

Description

Segment a single indel sequence using Rcpp interface

Usage

```
seg_simple(ins_or_del, string, context)
```

Arguments

ins_or_del	Character indicating insertion ("i") or deletion ("d")
string	A single indel sequence to segment (character scalar)
context	A single flanking context sequence (character scalar)

Details

This function segments an indel sequence by finding the optimal repeat unit that best explains the sequence structure. The algorithm tries all possible repeat unit sizes and selects the best segmentation based on:

- Highest 3' flanking repeat count (most important)
- Highest internal repeat count
- Lowest spacer length (prefer clean repeats)
- Lowest unit length (prefer simpler units)

Value

A list with the following elements:

unit	Repeat unit sequence (character)
unit_length	Unit length (integer)
internal_rep	Internal repeat region (character)
internal_reps	Internal repeat count (integer)
spacer	Spacer sequence (character)
spacer_length	Spacer length (integer)
prime3_rep	3' flanking repeat region (character)
prime3_reps	3' flanking repeat count (integer)
original_reps	Original repeat count (integer)

Examples

```
## Not run:
# Simple AT repeat
result <- seg_simple("d", "ATATAT", "ATATGG")
print(result)

# CG repeat
result <- seg_simple("d", "CGCGCG", "CGCGAA")
print(result)

## End(Not run)
```

SimpleReadVCF

Read a VCF file into a data frame with minimal processing.

Description

Read a VCF file into a data frame with minimal processing.

Usage

```
SimpleReadVCF(file)
```


Arguments

`file` The name/path of the VCF file, or a complete URL.

Details

Header lines beginning "##" are removed, and column "#CHROM" is renamed to "CHROM". Other column names are unchanged. Columns "#CHROM", "POS", "REF", and "ALT" must be in the input.

Value

A data frame storing mutation records of a VCF file.

Examples

```
file <- c(system.file("extdata/Strelka-SBS-vcf",
                     "Strelka.SBS.GRCh37.s1.vcf",
                     package = "ICAMS"))
df <- SimpleReadVCF(file)
```

SplitListOfVCFs	<i>Split each VCF into SBS, DBS, and ID VCFs (plus VCF-like data frame with left-over rows)</i>
-----------------	---

Description

Split each VCF into SBS, DBS, and ID VCFs (plus VCF-like data frame with left-over rows)

Usage

```
SplitListOfVCFs(
  list.of.vcfs,
  variant.caller,
  max.vaf.diff = 0.02,
  num.of.cores = 1,
  suppress.discarded.variants.warnings = TRUE,
  always.merge.SBS = FALSE,
  chr.names.to.process = NULL
)
```

Arguments

<code>list.of.vcfs</code>	List of VCFs as in-memory data frames. The VCFs should have VAF and <code>read.depth</code> information added. See <code>ReadVCFs</code> for more details.
<code>variant.caller</code>	Name of the variant caller that produces the VCF, can be either "strelka", "mutect", "freebayes" or "unknown". If variant caller is "mutect", do not merge SBSs into DBS.
<code>max.vaf.diff</code>	The maximum difference of VAF, default value is 0.02. If the absolute difference of VAFs for adjacent SBSs is bigger than <code>max.vaf.diff</code> , then these adjacent SBSs are likely to be "merely" asynchronous single base mutations, opposed to a simultaneous doublet mutation or variants involving more than two consecutive bases. Use negative value (e.g. -1) to suppress merging adjacent SBSs to DBS.

num.of.cores The number of cores to use. Not available on Windows unless num.of.cores = 1.

suppress.discarded.variants.warnings Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

always.merge.SBS If TRUE merge adjacent SBSs as DBSs regardless of VAFs and regardless of the value of max.vaf.diff. It is an error to set this to TRUE when variant.caller = "mutect".

chr.names.to.process A character vector specifying the chromosome names in VCF whose variants will be kept and processed, other chromosome variants will be discarded. If NULL (default), all variants will be kept except those on chromosomes with names that contain strings "GL", "KI", "random", "Hs", "M", "JH", "fix", "alt".

Value

A list containing the following objects:

- SBS: List of VCFs with only single base substitutions.
- DBS: List of VCFs with only doublet base substitutions as called by Mutect.
- ID: List of VCFs with only small insertions and deletions.
- discarded.variants: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column discarded.reason for more details.

Examples

```
file <- c(system.file("extdata/Mutect-vcf",
                     "Mutect.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.vcfs <- ReadVCFs(file, variant.caller = "mutect")
split.vcfs <- SplitListOfVCFs(list.of.vcfs, variant.caller = "mutect")
```

StrelkaIDVCFFilesToCatalog

[Deprecated, use VCFsToCatalogs(variant.caller = "strelka") instead] Create ID (small insertions and deletions) catalog from Strelka ID VCF files

Description

[Deprecated, use VCFsToCatalogs(variant.caller = "strelka") instead] Create ID (small insertions and deletions) catalog from the Strelka ID VCFs specified by files

Usage

```
StrelkaIDVCFFilesToCatalog(
  files,
  ref.genome,
  region = "unknown",
  names.of.VCFs = NULL,
```

```

    flag.mismatches = 0,
    return.annotated.vcfs = FALSE,
    suppress.discarded.variants.warnings = TRUE
)

```

Arguments

<code>files</code>	Character vector of file paths to the Strelka ID VCF files.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>region</code>	A character string designating a genomic region; see as.catalog and ICAMS .
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCF file paths in <code>files</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>files</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
<code>flag.mismatches</code>	Deprecated. If there are ID variants whose REF do not match the extracted sequence from <code>ref.genome</code> , the function will automatically discard these variants and an element <code>discarded.variants</code> will appear in the return value. See AnnotateIDVCF for more details.
<code>return.annotated.vcfs</code>	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is <code>FALSE</code> .
<code>suppress.discarded.variants.warnings</code>	Logical. Whether to suppress warning messages showing information about the discarded variants. Default is <code>TRUE</code> .

Details

This function calls [VCFsToIDCatalogs](#)

Value

A **list** of elements:

- `catalog`: The ID (small insertions and deletions) catalog with attributes added. See [as.catalog](#) for more details.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of data frames which contain the original VCF's ID mutation rows with three additional columns `seq.context.width`, `seq.context` and `ID.class` added. The category assignment of each ID mutation in VCF can be obtained from `ID.class` column.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Examples

```
## Not run:
file <- c(system.file("extdata/Strelka-ID-vcf",
                     "Strelka.ID.GRCh37.s1.vcf",
                     package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catID <- StrelkaIDVCFFilesToCatalog(file, ref.genome = "hg19",
                                     region = "genome")
}

## End(Not run)
```

StrelkaIDVCFFilesToCatalogAndPlotToPdf

Create ID (small insertions and deletions) catalog from Strelka ID VCF files and plot them to PDF

Description

Deprecated, use `VCFsToCatalogsAndPlotToPdf(variant.caller = "strelka")` instead Create ID (small insertions and deletions) catalog from the Strelka ID VCFs specified by files and plot them to PDF

Usage

```
StrelkaIDVCFFilesToCatalogAndPlotToPdf(
  files,
  ref.genome,
  region = "unknown",
  names.of.VCFs = NULL,
  output.file = "",
  flag.mismatches = 0,
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

<code>files</code>	Character vector of file paths to the Strelka ID VCF files.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>region</code>	A character string designating a genomic region; see as.catalog and ICAMS .
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCF file paths in <code>files</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>files</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.

`output.file` Optional. The base name of the PDF file to be produced; the file is ending in `catID.pdf`.

`flag.mismatches` Deprecated. If there are ID variants whose REF do not match the extracted sequence from `ref.genome`, the function will automatically discard these variants and an element `discarded.variants` will appear in the return value. See [AnnotateIDVCF](#) for more details.

`return.annotated.vcfs` Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.

`suppress.discarded.variants.warnings` Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

Details

This function calls [StrelkaIDVCFFilesToCatalog](#) and [PlotCatalogToPdf](#)

Value

A **list** of elements:

- `catalog`: The ID (small insertions and deletions) catalog with attributes added. See [as.catalog](#) for more details.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of data frames which contain the original VCF's ID mutation rows with three additional columns `seq.context.width`, `seq.context` and `ID.class` added. The category assignment of each ID mutation in VCF can be obtained from `ID.class` column.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Examples

```
## Not run:
file <- c(system.file("extdata/Strelka-ID-vcf",
                     "Strelka.ID.GRCh37.s1.vcf",
                     package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catID <-
```

```

StrelkaIDVCFFilesToCatalogAndPlotToPdf(file, ref.genome = "hg19",
                                         region = "genome",
                                         output.file =
                                         file.path(tempdir(), "StrelkaID"))}

## End(Not run)

```

StrelkaIDVCFFilesToZipFile

Create a zip file which contains ID (small insertions and deletions) catalog and plot PDF from Strelka ID VCF files

Description

Deprecated, use `VCFsToZipFile(variant.caller = "strelka")` instead Create ID (small insertions and deletions) catalog from the Strelka ID VCFs specified by `dir`, save the catalog as CSV file, plot it to PDF and generate a zip archive of all the output files.

Usage

```

StrelkaIDVCFFilesToZipFile(
  dir,
  zipfile,
  ref.genome,
  region = "unknown",
  names.of.VCFs = NULL,
  base.filename = "",
  flag.mismatches = 0,
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)

```

Arguments

<code>dir</code>	Pathname of the directory which contains only the Strelka ID VCF files. Each Strelka ID VCF must have a file extension ".vcf" (case insensitive) and share the same <code>ref.genome</code> and region.
<code>zipfile</code>	Pathname of the zip file to be created.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>region</code>	A character string designating a genomic region; see as.catalog and ICAMS .
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCFs listed in <code>dir</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>dir</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
<code>base.filename</code>	Optional. The base name of the CSV and PDF file to be produced; the file is ending in <code>catID.csv</code> and <code>catID.pdf</code> respectively.

flag.mismatches

Deprecated. If there are ID variants whose REF do not match the extracted sequence from `ref.genome`, the function will automatically discard these variants and an element `discarded.variants` will appear in the return value. See [AnnotateIDVCF](#) for more details.

return.annotated.vcfs

Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.

suppress.discarded.variants.warnings

Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

Details

This function calls [StrelkaIDVCFFilesToCatalog](#), [PlotCatalogToPdf](#), [WriteCatalog](#) and `zip::zipr`.

Value

A list of elements:

- `catalog`: The ID (small insertions and deletions) catalog with attributes added. See [as.catalog](#) for more details.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of data frames which contain the original VCF's ID mutation rows with three additional columns `seq.context.width`, `seq.context` and `ID.class` added. The category assignment of each ID mutation in VCF can be obtained from `ID.class` column.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Examples

```
## Not run:
dir <- c(system.file("extdata/Strelka-ID-vcf",
  package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs <-
    StrelkaIDVCFFilesToZipFile(dir,
      zipfile = file.path(tempdir(), "test.zip"),
      ref.genome = "hg19",
      region = "genome",
```

```

                                base.filename = "Strelka-ID")
  unlink(file.path(tempdir(), "test.zip"))}

## End(Not run)

```

StrelkaSBSVCFFilesToCatalog

Create SBS and DBS catalogs from Strelka SBS VCF files

Description

Deprecated, use `VCFsToCatalogs(variant.caller = "strelka")` instead Create 3 SBS catalogs (96, 192, 1536) and 3 DBS catalogs (78, 136, 144) from the Strelka SBS VCFs specified by files. The function will find and merge adjacent SBS pairs into DBS if their VAFs are very similar. The default threshold value for VAF is 0.02.

Usage

```

StrelkaSBSVCFFilesToCatalog(
  files,
  ref.genome,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)

```

Arguments

<code>files</code>	Character vector of file paths to the Strelka SBS VCF files.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>trans.ranges</code>	Optional. If <code>ref.genome</code> specifies one of the BSgenome object <code>BSgenome.Hsapiens.1000genomes.hs37d5</code> <code>BSgenome.Hsapiens.UCSC.hg38</code> <code>BSgenome.Mmusculus.UCSC.mm10</code> then the function will infer <code>trans.ranges</code> automatically. Otherwise, user will need to provide the necessary <code>trans.ranges</code> . Please refer to TranscriptRanges for more details. If <code>is.null(trans.ranges)</code> do not add transcript range information.
<code>region</code>	A character string designating a genomic region; see as.catalog and ICAMS .
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCF file paths in <code>files</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>files</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
<code>return.annotated.vcfs</code>	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is <code>FALSE</code> .
<code>suppress.discarded.variants.warnings</code>	Logical. Whether to suppress warning messages showing information about the discarded variants. Default is <code>TRUE</code> .

Value

A list containing the following objects:

```
## Not run:  
file <- c(system.file("extdata/Strelka-SBS-vcf",  
                      "Strelka.SBS.GRCh37.s1.vcf",  
                      package = "ICAMS"))  
  
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {  
  catalogs <- StrelkaSBSVCFFilesToCatalog(file, ref.genome = "hg19",  
                                           trans.ranges = trans.ranges.GRCh37,  
                                           region = "genome")}  
  
## End(Not run)
```

StrelkaSBSVCFFilesToCatalogAndPlotToPdf

[Deprecated, use `VCFsToCatalogsAndPlotToPdf(variant.caller = "strelka")` instead] Create SBS and DBS catalogs from Strelka SBS VCF files and plot them to PDF

Description

[Deprecated, use `VCFsToCatalogsAndPlotToPdf(variant.caller = "strelka")` instead] Create 3 SBS catalogs (96, 192, 1536) and 3 DBS catalogs (78, 136, 144) from the Strelka SBS VCFs specified by `files` and plot them to PDF. The function will find and merge adjacent SBS pairs into DBS if their VAFs are very similar. The default threshold value for VAF is 0.02.

Usage

```
StrelkaSBSVCFFilesToCatalogAndPlotToPdf(
  files,
  ref.genome,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  output.file = "",
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

<code>files</code>	Character vector of file paths to the Strelka SBS VCF files.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>trans.ranges</code>	Optional. If <code>ref.genome</code> specifies one of the BSgenome object <code>BSgenome.Hsapiens.1000genomes.hs37d5</code> <code>BSgenome.Hsapiens.UCSC.hg38</code> <code>BSgenome.Mmusculus.UCSC.mm10</code> then the function will infer <code>trans.ranges</code> automatically. Otherwise, user will need to provide the necessary <code>trans.ranges</code> . Please refer to TranscriptRanges for more details. If <code>is.null(trans.ranges)</code> do not add transcript range information.
<code>region</code>	A character string designating a genomic region; see as.catalog and ICAMS .
<code>names.of.VCFs</code>	Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCF file paths in <code>files</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>files</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
<code>output.file</code>	Optional. The base name of the PDF files to be produced; multiple files will be generated, each ending in <code>x.pdf</code> , where <code>x</code> indicates the type of catalog plotted in the file.

```
return.annotated.vcfs
```

Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.

```
suppress.discarded.variants.warnings
```

Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

Details

This function calls [StrelkaSBSVCFFilesToCatalog](#) and [PlotCatalogToPdf](#)

Value

A list containing the following objects:

- `catSBS96`, `catSBS192`, `catSBS1536`: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- `catDBS78`, `catDBS136`, `catDBS144`: Matrix of 3 DBS catalogs (one each for 78, 136, and 144).
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of elements:
 - SBS: SBS VCF annotated by [AnnotateSBSVCF](#) with three new columns `SBS96.class`, `SBS192.class` and `SBS1536.class` showing the mutation class for each SBS variant.
 - DBS: DBS VCF annotated by [AnnotateDBSVCF](#) with three new columns `DBS78.class`, `DBS136.class` and `DBS144.class` showing the mutation class for each DBS variant.

If `trans.ranges` is not provided by user and cannot be inferred by ICAMS, SBS 192 and DBS 144 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

Note

SBS 192 and DBS 144 catalogs include only mutations in transcribed regions.

Comments

To add or change attributes of the catalog, you can use function [attr](#).
For example, `attr(catalog, "abundance") <- custom.abundance`.

Examples

```
## Not run:
file <- c(system.file("extdata/Strelka-SBS-vcf",
                     "Strelka.SBS.GRCh37.s1.vcf",
                     package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs <-
    StrelkaSBSVCFFilesToCatalogAndPlotToPdf(file, ref.genome = "hg19",
                                             trans.ranges = trans.ranges.GRCh37,
                                             region = "genome",
                                             output.file =
                                             file.path(tempdir(), "StrelkaSBS"))}

## End(Not run)
```

StrelkaSBSVCFFilesToZipFile

Create a zip file which contains catalogs and plot PDFs from Strelka SBS VCF files

Description

Deprecated, use `VCFsToZipFile(variant.caller = "strelka")` instead Create 3 SBS catalogs (96, 192, 1536), 3 DBS catalogs (78, 136, 144) from the Strelka SBS VCFs specified by `dir`, save the catalogs as CSV files, plot them to PDF and generate a zip archive of all the output files. The function will find and merge adjacent SBS pairs into DBS if their VAFs are very similar. The default threshold value for VAF is 0.02.

Usage

```
StrelkaSBSVCFFilesToZipFile(
  dir,
  zipfile,
  ref.genome,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  base.filename = "",
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

- | | |
|----------------------------|--|
| <code>dir</code> | Pathname of the directory which contains only the Strelka SBS VCF files. Each Strelka SBS VCF must have a file extension ".vcf" (case insensitive) and share the same <code>ref.genome</code> and region. |
| <code>zipfile</code> | Pathname of the zip file to be created. |
| <code>ref.genome</code> | A <code>ref.genome</code> argument as described in ICAMS . |
| <code>trans.ranges</code> | Optional. If <code>ref.genome</code> specifies one of the BSgenome object <ol style="list-style-type: none"> 1. <code>BSgenome.Hsapiens.1000genomes.hs37d5</code> 2. <code>BSgenome.Hsapiens.UCSC.hg38</code> 3. <code>BSgenome.Mmusculus.UCSC.mm10</code> then the function will infer <code>trans.ranges</code> automatically. Otherwise, user will need to provide the necessary <code>trans.ranges</code> . Please refer to TranscriptRanges for more details. If <code>is.null(trans.ranges)</code> do not add transcript range information. |
| <code>region</code> | A character string designating a genomic region; see as.catalog and ICAMS . |
| <code>names.of.VCFs</code> | Optional. Character vector of names of the VCF files. The order of names in <code>names.of.VCFs</code> should match the order of VCFs listed in <code>dir</code> . If <code>NULL</code> (default), this function will remove all of the path up to and including the last path separator (if any) in <code>dir</code> and file paths without extensions (and the leading dot) will be used as the names of the VCF files. |

`base.filename` Optional. The base name of the CSV and PDF files to be produced; multiple files will be generated, each ending in `x.csv` or `x.pdf`, where `x` indicates the type of catalog.

`return.annotated.vcfs` Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.

`suppress.discarded.variants.warnings` Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

Details

This function calls [StrelkaSBSVCFFilesToCatalog](#), [PlotCatalogToPdf](#), [WriteCatalog](#) and `zip::zipr`.

Value

A list containing the following objects:

- `catSBS96`, `catSBS192`, `catSBS1536`: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- `catDBS78`, `catDBS136`, `catDBS144`: Matrix of 3 DBS catalogs (one each for 78, 136, and 144).
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of elements:
 - SBS: SBS VCF annotated by [AnnotateSBSVCF](#) with three new columns `SBS96.class`, `SBS192.class` and `SBS1536.class` showing the mutation class for each SBS variant.
 - DBS: DBS VCF annotated by [AnnotateDBSVCF](#) with three new columns `DBS78.class`, `DBS136.class` and `DBS144.class` showing the mutation class for each DBS variant.

If `trans.ranges` is not provided by user and cannot be inferred by ICAMS, SBS 192 and DBS 144 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

Note

SBS 192 and DBS 144 catalogs include only mutations in transcribed regions.

Comments

To add or change attributes of the catalog, you can use function [attr](#).
For example, `attr(catalog, "abundance") <- custom.abundance`.

Examples

```
## Not run:
dir <- c(system.file("extdata/Strelka-SBS-vcf",
  package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs <-
    StrelkaSBSVCFFilesToZipFile(dir,
      zipfile = file.path(tempdir(), "test.zip"),
      ref.genome = "hg19",
      trans.ranges = trans.ranges.GRCh37,
```

```

        region = "genome",
        base.filename = "Strelka-SBS")
unlink(file.path(tempdir(), "test.zip"))

## End(Not run)

```

TranscriptRanges	<i>Transcript ranges data</i>
------------------	-------------------------------

Description

Transcript ranges and strand information for a particular reference genome.

Usage

```
trans.ranges.GRCh37
```

```
trans.ranges.GRCh38
```

```
trans.ranges.GRCm38
```

Format

A [data.table](#) which contains transcript range and strand information for a particular reference genome. colnames are chrom, start, end, strand, Ensembl.gene.ID, gene.symbol. It uses one-based coordinates.

An object of class `data.table` (inherits from `data.frame`) with 19083 rows and 6 columns.

An object of class `data.table` (inherits from `data.frame`) with 19096 rows and 6 columns.

An object of class `data.table` (inherits from `data.frame`) with 20325 rows and 6 columns.

Details

This information is needed to generate catalogs that depend on transcriptional strand information, for example catalogs of class `SBS192Catalog`.

`trans.ranges.GRCh37`: **Human** GRCh37.

`trans.ranges.GRCh38`: **Human** GRCh38.

`trans.ranges.GRCm38`: **Mouse** GRCm38.

For these two tables, only genes that are associated with a CCDS ID are kept for transcriptional strand bias analysis.

This information is needed for [StrelkaSBSVCFFilesToCatalog](#), [StrelkaSBSVCFFilesToCatalogAndPlotToPdf](#), [MutectVCFFilesToCatalog](#), [MutectVCFFilesToCatalogAndPlotToPdf](#), [VCFsToSBSCatalogs](#) and [VCFsToDBSCatalogs](#).

Source

ftp://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_30/GRCh37_mapping/gencode.v30lift37.annotation.gff3.gz

ftp://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_30/gencode.v30.annotation.gff3.gz

ftp://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_mouse/release_M21/gencode.vM21.annotation.gff3.gz

Examples

```
trans.ranges.GRCh37
# chrom      start      end strand Ensembl.gene.ID  gene.symbol
#      1      65419      71585      + ENSG00000186092    OR4F5
#      1     367640     368634      + ENSG00000235249    OR4F29
#      1     621059     622053      - ENSG00000284662    OR4F16
#      1     859308     879961      + ENSG00000187634    SAMD11
#      1     879583     894689      - ENSG00000188976    NOC2L
#      ...           ...           ...           ...           ...           ...
```

TransformCatalog	<i>Transform between counts and density spectrum catalogs and counts and density signature catalogs</i>
------------------	---

Description

Transform between counts and density spectrum catalogs and counts and density signature catalogs

Usage

```
TransformCatalog(
  catalog,
  target.ref.genome = NULL,
  target.region = NULL,
  target.catalog.type = NULL,
  target.abundance = NULL
)
```

Arguments

catalog	An SBS or DBS catalog as described in ICAMS ; must not be an ID (small insertions and deletions) catalog.
target.ref.genome	A ref.genome argument as described in ICAMS . If NULL, then defaults to the ref.genome attribute of catalog.
target.region	A region argument; see as.catalog and ICAMS . If NULL, then defaults to the region attribute of catalog.
target.catalog.type	A character string acting as a catalog type identifier, one of "counts", "density", "counts.signature", "density.signature"; see as.catalog . If NULL, then defaults to the catalog.type attribute of catalog.
target.abundance	A vector of counts, one for each source K-mer for mutations (e.g. for strand-agnostic single nucleotide substitutions in trinucleotide – i.e. 3-mer – context, one count each for ACA, ACC, ACG, ... TTT). See all.abundance . If NULL, the function tries to infer target.abundance from the class of catalog and the value of the target.ref.genome, target.region, and target.catalog.type.

Details

Only the following transformations are legal:

1. counts -> counts (deprecated, generates a warning; we strongly suggest that you work with densities if comparing spectra or signatures generated from data with different underlying abundances.)
2. counts -> density
3. counts -> (counts.signature, density.signature)
4. density -> counts (the semantics are to infer the genome-wide or exome-wide counts based on the densities)
5. density -> density (a null operation, generates a warning)
6. density -> (counts.signature, density.signature)
7. counts.signature -> counts.signature (used to transform between the source abundance and target.abundance)
8. counts.signature -> density.signature
9. counts.signature -> (counts, density) (generates an error)
10. density.signature -> density.signature (a null operation, generates a warning)
11. density.signature -> counts.signature
12. density.signature -> (counts, density) (generates an error)

Value

A catalog as defined in [ICAMS](#).

Rationale

The [TransformCatalog](#) function transforms catalogs of mutational spectra or signatures to account for differing abundances of the source sequence of the mutations in the genome.

For example, mutations from ACG are much rarer in the human genome than mutations from ACC simply because CG dinucleotides are rare in the genome. Consequently, there are two possible representations of mutational spectra or signatures. One representation is based on mutation counts as observed in a given genome or exome, and this approach is widely used, as, for example, at <https://cancer.sanger.ac.uk/cosmic/signatures>, which presents signatures based on observed mutation counts in the human genome. We call these "counts-based spectra" or "counts-based signatures".

Alternatively, mutational spectra or signatures can be represented as mutations per source sequence, for example the number of ACT > AGT mutations occurring at all ACT 3-mers in a genome. We call these "density-based spectra" or "density-based signatures".

This function can also transform spectra based on observed genome-wide counts to "density"-based catalogs. In density-based catalogs mutations are expressed as mutations per source sequences. For example, a density-based catalog represents the proportion of ACCs mutated to ATCs, the proportion of ACGs mutated to ATGs, etc. This is different from counts-based mutational spectra catalogs, which contain the number of ACC > ATC mutations, the number of ACG > ATG mutations, etc.

This function can also transform observed-count based spectra or signatures from genome to exome based counts, or between different species (since the abundances of source sequences vary between genome and exome and between species).

Examples

```
file <- system.file("extdata",
  "strelka.regress.cat.sbs.96.csv",
  package = "ICAMS")
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catSBS96.counts <- ReadCatalog(file, ref.genome = "hg19",
    region = "genome",
    catalog.type = "counts")
  catSBS96.density <- TransformCatalog(catSBS96.counts,
    target.ref.genome = "hg19",
    target.region = "genome",
    target.catalog.type = "density")}
```

VCFsToCatalogs

Create SBS, DBS and Indel catalogs from VCFs

Description

Create 3 SBS catalogs (96, 192, 1536), 3 DBS catalogs (78, 136, 144) and Indel catalog from the Mutect VCFs specified by files

Usage

```
VCFsToCatalogs(
  files,
  ref.genome,
  variant.caller = "unknown",
  num.of.cores = 1,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  tumor.col.names = NA,
  filter.status = DefaultFilterStatus(variant.caller),
  get.vaf.function = NULL,
  ...,
  max.vaf.diff = 0.02,
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE,
  chr.names.to.process = NULL
)
```

Arguments

<code>files</code>	Character vector of file paths to the VCF files.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>variant.caller</code>	Name of the variant caller that produces the VCF, can be either "strelka", "mutect", "freebayes" or "unknown". This information is needed to calculate the VAFs (variant allele frequencies). If variant caller is "unknown" (default) and <code>get.vaf.function</code> is NULL, then VAF and read depth will be NAs. If variant caller is "mutect", do not merge SBSs into DBS.

num.of.cores	The number of cores to use. Not available on Windows unless num.of.cores = 1.
trans.ranges	Optional. If ref.genome specifies one of the BSgenome object 1. BSgenome.Hsapiens.1000genomes.hs37d5 2. BSgenome.Hsapiens.UCSC.hg38 3. BSgenome.Mmusculus.UCSC.mm10 then the function will infer trans.ranges automatically. Otherwise, user will need to provide the necessary trans.ranges. Please refer to TranscriptRanges for more details. If is.null(trans.ranges) do not add transcript range information.
region	A character string designating a genomic region; see as.catalog and ICAMS .
names.of.VCFs	Optional. Character vector of names of the VCF files. The order of names in names.of.VCFs should match the order of VCF file paths in files. If NULL(default), this function will remove all of the path up to and including the last path separator (if any) in files and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
tumor.col.names	Optional. Only applicable to Mutect VCFs. Vector of column names or column indices in Mutect VCFs which contain the tumor sample information. The order of elements in tumor.col.names should match the order of Mutect VCFs specified in files. If tumor.col.names is equal to NA(default), this function will use the 10th column in all the Mutect VCFs to calculate VAFs. See GetMutectVAF for more details.
filter.status	The character string in column FILTER of the VCF that indicates that a variant has passed all the variant caller's filters. Variants (lines in the VCF) for which the value in column FILTER does not equal filter.status are silently excluded from the output. The internal function DefaultFilterStatus tries to infer filter.status based on variant.caller. If variant.caller is "unknown", user must specify filter.status explicitly. If filter.status = NULL, all variants are retained. If there is no FILTER column in the VCF, all variants are retained with a warning.
get.vaf.function	Optional. Only applicable when variant.caller is " unknown ". Function to calculate VAF(variant allele frequency) and read depth information from original VCF. See GetMutectVAF as an example. If NULL(default) and variant.caller is "unknown", then VAF and read depth will be NAs.
...	Optional arguments to get.vaf.function.
max.vaf.diff	Not applicable if variant.caller = "mutect". The maximum difference of VAF, default value is 0.02. If the absolute difference of VAFs for adjacent SBSs is bigger than max.vaf.diff, then these adjacent SBSs are likely to be "merely" asynchronous single base mutations, opposed to a simultaneous doublet mutation or variants involving more than two consecutive bases. Use negative value (e.g. -1) to suppress merging adjacent SBSs to DBS.
return.annotated.vcfs	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.
suppress.discarded.variants.warnings	Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

`chr.names.to.process`

A character vector specifying the chromosome names in VCF whose variants will be kept and processed, other chromosome variants will be discarded. If NULL(default), all variants will be kept except those on chromosomes with names that contain strings "GL", "KI", "random", "Hs", "M", "JH", "fix", "alt".

Details

This function calls [VCFsToSBSCatalogs](#), [VCFsToDBSCatalogs](#) and [VCFsToIDCatalogs](#)

Value

A list containing the following objects:

- `catSBS96`, `catSBS192`, `catSBS1536`: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- `catDBS78`, `catDBS136`, `catDBS144`: Matrix of 3 DBS catalogs (one each for 78, 136, and 144).
- `catID`: Matrix of ID (small insertions and deletions) catalog.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of elements:
 - SBS: SBS VCF annotated by [AnnotateSBSVCF](#) with three new columns `SBS96.class`, `SBS192.class` and `SBS1536.class` showing the mutation class for each SBS variant.
 - DBS: DBS VCF annotated by [AnnotateDBSVCF](#) with three new columns `DBS78.class`, `DBS136.class` and `DBS144.class` showing the mutation class for each DBS variant.
 - ID: ID VCF annotated by [AnnotateIDVCF](#) with one new column `ID.class` showing the mutation class for each ID variant.

If `trans.ranges` is not provided by user and cannot be inferred by ICAMS, SBS 192 and DBS 144 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

SBS 192 and DBS 144 catalogs include only mutations in transcribed regions. In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Comments

To add or change attributes of the catalog, you can use function [attr](#). For example, `attr(catalog, "abundance") <- custom.abundance`.

Examples

```
file <- c(system.file("extdata/Mutect-vcf",
                     "Mutect.GRCh37.s1.vcf",
                     package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs <- VCFsToCatalogs(file, ref.genome = "hg19",
                             variant.caller = "mutect", region = "genome")}
```

VCFsToCatalogsAndPlotToPdf

Create SBS, DBS and Indel catalogs from VCFs and plot them to PDF

Description

Create 3 SBS catalogs (96, 192, 1536), 3 DBS catalogs (78, 136, 144) and Indel catalog from the VCFs specified by files and plot them to PDF

Usage

```
VCFsToCatalogsAndPlotToPdf(
  files,
  output.dir,
  ref.genome,
  variant.caller = "unknown",
  num.of.cores = 1,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  tumor.col.names = NA,
  filter.status = DefaultFilterStatus(variant.caller),
  get.vaf.function = NULL,
  ...,
  max.vaf.diff = 0.02,
  base.filename = "",
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE,
  chr.names.to.process = NULL
)
```

Arguments

<code>files</code>	Character vector of file paths to the VCF files.
<code>output.dir</code>	The directory where the PDF files will be saved.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>variant.caller</code>	Name of the variant caller that produces the VCF, can be either "strelka", "mutect", "freebayes" or "unknown". This information is needed to calculate the VAFs (variant allele frequencies). If variant caller is "unknown" (default) and <code>get.vaf.function</code> is NULL, then VAF and read depth will be NAs. If variant caller is "mutect", do not merge SBSs into DBS.

num.of.cores	The number of cores to use. Not available on Windows unless num.of.cores = 1.
trans.ranges	Optional. If ref.genome specifies one of the BSgenome object 1. BSgenome.Hsapiens.1000genomes.hs37d5 2. BSgenome.Hsapiens.UCSC.hg38 3. BSgenome.Mmusculus.UCSC.mm10 then the function will infer trans.ranges automatically. Otherwise, user will need to provide the necessary trans.ranges. Please refer to TranscriptRanges for more details. If is.null(trans.ranges) do not add transcript range information.
region	A character string designating a genomic region; see as.catalog and ICAMS .
names.of.VCFs	Optional. Character vector of names of the VCF files. The order of names in names.of.VCFs should match the order of VCF file paths in files. If NULL(default), this function will remove all of the path up to and including the last path separator (if any) in files and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
tumor.col.names	Optional. Only applicable to Mutect VCFs. Vector of column names or column indices in Mutect VCFs which contain the tumor sample information. The order of elements in tumor.col.names should match the order of Mutect VCFs specified in files. If tumor.col.names is equal to NA(default), this function will use the 10th column in all the Mutect VCFs to calculate VAFs. See GetMutectVAF for more details.
filter.status	The character string in column FILTER of the VCF that indicates that a variant has passed all the variant caller's filters. Variants (lines in the VCF) for which the value in column FILTER does not equal filter.status are silently excluded from the output. The internal function DefaultFilterStatus tries to infer filter.status based on variant.caller. If variant.caller is "unknown", user must specify filter.status explicitly. If filter.status = NULL, all variants are retained. If there is no FILTER column in the VCF, all variants are retained with a warning.
get.vaf.function	Optional. Only applicable when variant.caller is " unknown ". Function to calculate VAF(variant allele frequency) and read depth information from original VCF. See GetMutectVAF as an example. If NULL(default) and variant.caller is "unknown", then VAF and read depth will be NAs.
...	Optional arguments to get.vaf.function.
max.vaf.diff	Not applicable if variant.caller = "mutect". The maximum difference of VAF, default value is 0.02. If the absolute difference of VAFs for adjacent SBSs is bigger than max.vaf.diff, then these adjacent SBSs are likely to be "merely" asynchronous single base mutations, opposed to a simultaneous doublet mutation or variants involving more than two consecutive bases. Use negative value (e.g. -1) to suppress merging adjacent SBSs to DBS.
base.filename	Optional. The base name of the PDF files to be produced; multiple files will be generated, each ending in x.pdf, where x indicates the type of catalog plotted in the file.
return.annotated.vcfs	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.

`suppress.discarded.variants.warnings`

Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

`chr.names.to.process`

A character vector specifying the chromosome names in VCF whose variants will be kept and processed, other chromosome variants will be discarded. If NULL(default), all variants will be kept except those on chromosomes with names that contain strings "GL", "KI", "random", "Hs", "M", "JH", "fix", "alt".

Details

This function calls [VCFsToCatalogs](#) and [PlotCatalogToPdf](#)

Value

A list containing the following objects:

- `catSBS96`, `catSBS192`, `catSBS1536`: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- `catDBS78`, `catDBS136`, `catDBS144`: Matrix of 3 DBS catalogs (one each for 78, 136, and 144).
- `catID`: Matrix of ID (small insertions and deletions) catalog.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of elements:
 - SBS: SBS VCF annotated by [AnnotateSBSVCF](#) with three new columns `SBS96.class`, `SBS192.class` and `SBS1536.class` showing the mutation class for each SBS variant.
 - DBS: DBS VCF annotated by [AnnotateDBSVCF](#) with three new columns `DBS78.class`, `DBS136.class` and `DBS144.class` showing the mutation class for each DBS variant.
 - ID: ID VCF annotated by [AnnotateIDVCF](#) with one new column `ID.class` showing the mutation class for each ID variant.

If `trans.ranges` is not provided by user and cannot be inferred by ICAMS, SBS 192 and DBS 144 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

SBS 192 and DBS 144 catalogs include only mutations in transcribed regions. In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Comments

To add or change attributes of the catalog, you can use function [attr](#).
For example, `attr(catalog, "abundance") <- custom.abundance`.

Examples

```
file <- c(system.file("extdata/Mutect-vcf",
                     "Mutect.GRCh37.s1.vcf",
                     package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs <-
    VCFsToCatalogsAndPlotToPdf(file, ref.genome = "hg19",
                               output.dir = tempdir(),
                               variant.caller = "mutect",
                               region = "genome",
                               base.filename = "Mutect")}
```

VCFsToDBSCatalogs	Create DBS catalogs from VCFs
-------------------	-------------------------------

Description

Create a list of 3 catalogs (one each for DBS78, DBS144 and DBS136) out of the contents in `list.of.DBS.vcfs`. The VCFs must not contain any type of mutation other than DBSs.

Usage

```
VCFsToDBSCatalogs(
  list.of.DBS.vcfs,
  ref.genome,
  num.of.cores = 1,
  trans.ranges = NULL,
  region = "unknown",
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

<code>list.of.DBS.vcfs</code>	List of in-memory data frames of pure DBS mutations – no SBS or 3+BS mutations. The list names will be the sample ids in the output catalog.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>num.of.cores</code>	The number of cores to use. Not available on Windows unless <code>num.of.cores = 1</code> .
<code>trans.ranges</code>	Optional. If <code>ref.genome</code> specifies one of the BSgenome object 1. <code>BSgenome.Hsapiens.1000genomes.hs37d5</code> 2. <code>BSgenome.Hsapiens.UCSC.hg38</code> 3. <code>BSgenome.Mmusculus.UCSC.mm10</code>

VCFsToIDCatalogs	Create ID (small insertions and deletions) catalog from ID VCFs
------------------	---

Description

Create ID (small insertions and deletions) catalog from ID VCFs

Usage

```
VCFsToIDCatalogs(
  list.of.vcfs,
  ref.genome,
  num.of.cores = 1,
  trans.ranges = NULL,
  region = "unknown",
  flag.mismatches = 0,
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

<code>list.of.vcfs</code>	List of in-memory ID VCFs. The list names will be the sample ids in the output catalog.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>num.of.cores</code>	The number of cores to use. Not available on Windows unless <code>num.of.cores = 1</code> .
<code>trans.ranges</code>	Optional. If <code>ref.genome</code> specifies one of the BSgenome object <code>BSgenome.Hsapiens.1000genomes.hs37d5</code> <code>BSgenome.Hsapiens.UCSC.hg38</code> <code>BSgenome.Mmusculus.UCSC.mm10</code> then the function will infer <code>trans.ranges</code> automatically. Otherwise, user will need to provide the necessary <code>trans.ranges</code> . Please refer to TranscriptRanges for more details. If <code>is.null(trans.ranges)</code> do not add transcript range information.
<code>region</code>	A character string acting as a region identifier, one of "genome", "exome".
<code>flag.mismatches</code>	Deprecated. If there are ID variants whose REF do not match the extracted sequence from <code>ref.genome</code> , the function will automatically discard these variants and an element <code>discarded.variants</code> will appear in the return value. See AnnotateIDVCF for more details.
<code>return.annotated.vcfs</code>	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.
<code>suppress.discarded.variants.warnings</code>	Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

Value

A **list** of elements:

- `catalog`: The ID (small insertions and deletions) catalog with attributes added. See [as.catalog](#) for details.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of data frames which contain the original VCF's ID mutation rows with three additional columns `seq.context.width`, `seq.context` and `ID.class` added. The category assignment of each ID mutation in VCF can be obtained from `ID.class` column.

Note

In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Examples

```
file <- c(system.file("extdata/Strelka-ID-vcf/",
                     "Strelka.ID.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.ID.vcfs <- ReadAndSplitVCFs(file, variant.caller = "strelka")$ID
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5",
                     quietly = TRUE)) {
  catID <- VCFsToIDCatalogs(list.of.ID.vcfs, ref.genome = "hg19",
                           region = "genome")}
```

VCFsToSBSCatalogs

Create SBS catalogs from SBS VCFs

Description

Create a list of 3 catalogs (one each for 96, 192, 1536) out of the contents in `list.of.SBS.vcfs`. The SBS VCFs must not contain DBSs, indels, or other types of mutations.

Usage

```
VCFsToSBSCatalogs(
  list.of.SBS.vcfs,
  ref.genome,
  num.of.cores = 1,
  trans.ranges = NULL,
  region = "unknown",
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE
)
```

Arguments

<code>list.of.SBS.vcfs</code>	List of in-memory data frames of pure SBS mutations – no DBS or 3+BS mutations. The list names will be the sample ids in the output catalog.
<code>ref.genome</code>	A <code>ref.genome</code> argument as described in ICAMS .
<code>num.of.cores</code>	The number of cores to use. Not available on Windows unless <code>num.of.cores = 1</code> .
<code>trans.ranges</code>	Optional. If <code>ref.genome</code> specifies one of the BSgenome object 1. <code>BSgenome.Hsapiens.1000genomes.hs37d5</code> 2. <code>BSgenome.Hsapiens.UCSC.hg38</code> 3. <code>BSgenome.Mmusculus.UCSC.mm10</code> then the function will infer <code>trans.ranges</code> automatically. Otherwise, user will need to provide the necessary <code>trans.ranges</code> . Please refer to TranscriptRanges for more details. If <code>is.null(trans.ranges)</code> do not add transcript range information.
<code>region</code>	A character string designating a genomic region; see as.catalog and ICAMS .
<code>return.annotated.vcfs</code>	Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.
<code>suppress.discarded.variants.warnings</code>	Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

Value

A list containing the following objects:

- `catSBS96`, `catSBS192`, `catSBS1536`: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. SBS VCF annotated by [AnnotateSBSVCF](#) with three new columns `SBS96.class`, `SBS192.class` and `SBS1536.class` showing the mutation class for each SBS variant.

If `trans.ranges` is not provided by user and cannot be inferred by [ICAMS](#), SBS 192 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

Comments

To add or change attributes of the catalog, you can use function `attr`.
For example, `attr(catalog, "abundance") <- custom.abundance`.

Note

SBS 192 catalogs only contain mutations in transcribed regions.

Examples

```
file <- c(system.file("extdata/Mutect-vcf",
                     "Mutect.GRCh37.s1.vcf",
                     package = "ICAMS"))
list.of.SBS.vcfs <- ReadAndSplitVCFs(file, variant.caller = "mutect")$SBS
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs.SBS <- VCFsToSBSCatalogs(list.of.SBS.vcfs, ref.genome = "hg19",
                                   trans.ranges = trans.ranges.GRCh37,
                                   region = "genome")}
```

VCFsToZipFile

Create a zip file which contains catalogs and plot PDFs from VCFs

Description

Create 3 SBS catalogs (96, 192, 1536), 3 DBS catalogs (78, 136, 144) and Indel catalog from the VCFs specified by `dir`, save the catalogs as CSV files, plot them to PDF and generate a zip archive of all the output files.

Usage

```
VCFsToZipFile(
  dir,
  files,
  zipfile,
  ref.genome,
  variant.caller = "unknown",
  num.of.cores = 1,
  trans.ranges = NULL,
  region = "unknown",
  names.of.VCFs = NULL,
  tumor.col.names = NA,
  filter.status = DefaultFilterStatus(variant.caller),
  get.vaf.function = NULL,
  ...,
  max.vaf.diff = 0.02,
  base.filename = "",
  return.annotated.vcfs = FALSE,
  suppress.discarded.variants.warnings = TRUE,
  chr.names.to.process = NULL
)
```

Arguments

dir	Pathname of the directory which contains VCFs that come from the same variant caller. Each VCF must have a file extension ".vcf" (case insensitive) and share the same ref.genome and region.
files	Character vector of file paths to the VCF files. Only one of argument dir or files need to be specified.
zipfile	Pathname of the zip file to be created.
ref.genome	A ref.genome argument as described in ICAMS .
variant.caller	Name of the variant caller that produces the VCF, can be either "strelka", "mutect", "freebayes" or "unknown". This information is needed to calculate the VAFs (variant allele frequencies). If variant caller is "unknown"(default) and get.vaf.function is NULL, then VAF and read depth will be NAs. If variant caller is "mutect", do not merge SBSs into DBS.
num.of.cores	The number of cores to use. Not available on Windows unless num.of.cores = 1.
trans.ranges	Optional. If ref.genome specifies one of the BSgenome object 1. BSgenome.Hsapiens.1000genomes.hs37d5 2. BSgenome.Hsapiens.UCSC.hg38 3. BSgenome.Mmusculus.UCSC.mm10 then the function will infer trans.ranges automatically. Otherwise, user will need to provide the necessary trans.ranges. Please refer to TranscriptRanges for more details. If is.null(trans.ranges) do not add transcript range information.
region	A character string designating a genomic region; see as.catalog and ICAMS .
names.of.VCFs	Optional. Character vector of names of the VCF files. The order of names in names.of.VCFs should match the order of VCF file paths in files. If NULL(default), this function will remove all of the path up to and including the last path separator (if any) in files and file paths without extensions (and the leading dot) will be used as the names of the VCF files.
tumor.col.names	Optional. Only applicable to Mutect VCFs. Vector of column names or column indices in Mutect VCFs which contain the tumor sample information. The order of elements in tumor.col.names should match the order of Mutect VCFs specified in files. If tumor.col.names is equal to NA(default), this function will use the 10th column in all the Mutect VCFs to calculate VAFs. See GetMutectVAF for more details.
filter.status	The character string in column FILTER of the VCF that indicates that a variant has passed all the variant caller's filters. Variants (lines in the VCF) for which the value in column FILTER does not equal filter.status are silently excluded from the output. The internal function DefaultFilterStatus tries to infer filter.status based on variant.caller. If variant.caller is "unknown", user must specify filter.status explicitly. If filter.status = NULL, all variants are retained. If there is no FILTER column in the VCF, all variants are retained with a warning.
get.vaf.function	Optional. Only applicable when variant.caller is " unknown ". Function to calculate VAF(variant allele frequency) and read depth information from original VCF. See GetMutectVAF as an example. If NULL(default) and variant.caller is "unknown", then VAF and read depth will be NAs.

... Optional arguments to `get.vaf.function`.

`max.vaf.diff` **Not** applicable if `variant.caller = "mutect"`. The maximum difference of VAF, default value is 0.02. If the absolute difference of VAFs for adjacent SBSs is bigger than `max.vaf.diff`, then these adjacent SBSs are likely to be "merely" asynchronous single base mutations, opposed to a simultaneous doublet mutation or variants involving more than two consecutive bases. Use negative value (e.g. -1) to suppress merging adjacent SBSs to DBS.

`base.filename` Optional. The base name of the CSV and PDF files to be produced; multiple files will be generated, each ending in `x.csv` or `x.pdf`, where `x` indicates the type of catalog.

`return.annotated.vcfs` Logical. Whether to return the annotated VCFs with additional columns showing mutation class for each variant. Default is FALSE.

`suppress.discarded.variants.warnings` Logical. Whether to suppress warning messages showing information about the discarded variants. Default is TRUE.

`chr.names.to.process` A character vector specifying the chromosome names in VCF whose variants will be kept and processed, other chromosome variants will be discarded. If NULL(default), all variants will be kept except those on chromosomes with names that contain strings "GL", "KI", "random", "Hs", "M", "JH", "fix", "alt".

Details

This function calls [VCFsToCatalogs](#), [PlotCatalogToPdf](#), [WriteCatalog](#) and `zip::zipr`.

Value

A list containing the following objects:

- `catSBS96`, `catSBS192`, `catSBS1536`: Matrix of 3 SBS catalogs (one each for 96, 192, and 1536).
- `catDBS78`, `catDBS136`, `catDBS144`: Matrix of 3 DBS catalogs (one each for 78, 136, and 144).
- `catID`: Matrix of ID (small insertions and deletions) catalog.
- `discarded.variants`: **Non-NULL only if** there are variants that were excluded from the analysis. See the added extra column `discarded.reason` for more details.
- `annotated.vcfs`: **Non-NULL only if** `return.annotated.vcfs = TRUE`. A list of elements:
 - SBS: SBS VCF annotated by [AnnotateSBSVCF](#) with three new columns `SBS96.class`, `SBS192.class` and `SBS1536.class` showing the mutation class for each SBS variant.
 - DBS: DBS VCF annotated by [AnnotateDBSVCF](#) with three new columns `DBS78.class`, `DBS136.class` and `DBS144.class` showing the mutation class for each DBS variant.
 - ID: ID VCF annotated by [AnnotateIDVCF](#) with one new column `ID.class` showing the mutation class for each ID variant.

If `trans.ranges` is not provided by user and cannot be inferred by ICAMS, SBS 192 and DBS 144 catalog will not be generated. Each catalog has attributes added. See [as.catalog](#) for more details.

ID classification

See https://github.com/steverozen/ICAMS/blob/v3.0.9-branch/data-raw/PCAWG7_indel_classification_2021_09_03.xlsx for additional information on ID (small insertions and deletions) mutation classification.

See the documentation for [Canonicalize1Del](#) which first handles deletions in homopolymers, then handles deletions in simple repeats with longer repeat units, (e.g. CACACACA, see [FindMaxRepeatDel](#)), and if the deletion is not in a simple repeat, looks for microhomology (see [FindDelMH](#)).

Note

SBS 192 and DBS 144 catalogs include only mutations in transcribed regions. In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Comments

To add or change attributes of the catalog, you can use function [attr](#).
For example, `attr(catalog, "abundance") <- custom.abundance`.

Examples

```
dir <- c(system.file("extdata/Mutect-vcf",
                    package = "ICAMS"))
if (requireNamespace("BSgenome.Hsapiens.1000genomes.hs37d5", quietly = TRUE)) {
  catalogs <-
    VCFsToZipFile(dir,
                  zipfile = file.path(tempdir(), "test.zip"),
                  ref.genome = "hg19",
                  variant.caller = "mutect",
                  region = "genome",
                  base.filename = "Mutect")
  unlink(file.path(tempdir(), "test.zip"))
}
```

WriteCatalog

Write a catalog

Description

Write a catalog to a file.

Usage

```
WriteCatalog(catalog, file, strict = TRUE)
```

Arguments

catalog	A catalog as defined in ICAMS ; see also as.catalog .
file	The path to the file to be created.
strict	If TRUE, do additional checks on the input, and stop if the checks fail.

Details

See also [ReadCatalog](#).

Note

In ID (small insertions and deletions) catalogs, deletion repeat sizes range from 0 to 5+, but for plotting and end-user documentation deletion repeat sizes range from 1 to 6+.

Examples

```
file <- system.file("extdata",  
                    "strelka.regress.cat.sbs.96.csv",  
                    package = "ICAMS")  
catSBS96 <- ReadCatalog(file)  
WriteCatalog(catSBS96, file = file.path(tempdir(), "catSBS96.csv"))
```


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